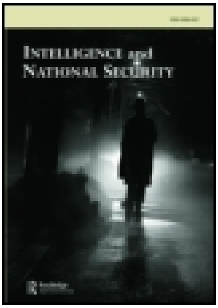

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Clausewitz, intelligence, uncertainty and the art of command in military operations

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This article explores the connections between intelligence, the conduct of military operations and the art of command. First, it examines this phenomenon in the period before the industrial revolution and the development of real-time communications, particularly as it was reflected in the influential arguments of Carl von Clausewitz. It also examines how far his ideas on intelligence and uncertainty shaped his theory of military operations. Second, the article studies the relationship between intelligence, operations and command in the modern period. For this purpose, it rests on a mature case study: Edward Drea's account of the use and misuse of intelligence in the Pacific War by General Douglas James MacArthur. Finally, the article considers the relationship between intelligence, operations and command in the future. It links together such issues as one general's use of Ultra and the interrelationship between intelligence, uncertainty and command in one war, and in the theory of war; it uses Clausewitz to illuminate a campaign and vice versa. The article cannot expand on every point it mentioned, but the hope is to provide a conceptual framework which may assist in their further development.

INTELLIGENCE AND MILITARY OPERATIONS

Any examination of intelligence and war must begin with Clausewitz's arguments about uncertainty and the psychology of command. Such a discussion may also illuminate those ideas which are fundamental to modern theories of war. The limits to this approach must be emphasized from the start, for both war and *On War* are complex matters. The discussion will therefore focus only on certain variables, excluding related issues such as technology, doctrine and politics.

Clausewitz believed that 'At the moment of battle, information about the strength of the enemy is uncertain, and the estimate of one's own is usually unrealistic'.[1] The inability to collect accurate, relevant and complete data about the battlefield, and on one's own forces no less than on the enemy, was fundamental to Clausewitz's view that war lies within

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'the realm of uncertainty . . . the realm of chance'. These views about intelligence were axiomatic but not simplistic. Clausewitz was referring not to man's ability to know the world but to a commander's ability to understand a battle. He was a practical man writing for practical men, through generalizations drawn from his own experience and study. He matured in a military environment where accurate operational intelligence was difficult to acquire, while rumors in battle were rife, where many important factors (perhaps most of them) were known to be unknown, where slow and unreliable communications systems rendered originally correct information inaccurate by the time it was received or, indeed, prevented it from ever being transmitted at all.

In the short period of time available to act on the battlefield, the command, communication and intelligence services of his era could not provide clear, consistent and accurate information. A general could not know the facts about operations that he needed to know when he needed to know them. He could not know the enemy or the battle: therefore he would have to act without knowledge.

From these specific circumstances Clausewitz derived universal generalizations, which were accurate for his time but not necessarily for ours. He concluded that

> Many intelligence reports in war are contradictory; even more are false and most are uncertain . . . most intelligence is false, and the effect of fear is to multiply lies and inaccuracies. As a rule most men would rather believe bad news than good, and rather tend to exaggerate the bad news. The dangers that are reported may soon, like waves, subside; but like waves they keep recurring, without apparent reason.

In these circumstances, more intelligence reports make 'us more, not less uncertain'.[2] As will be shown, that observation remains true even today, though the nature of the uncertainty has changed.

Some of this analysis was commonplace – soldiers had often emphasized the limited value of intelligence. Xenophon, for example, wrote that 'in time of war enemies often form designs on one another but seldom know the state of things among the party against whom their designs are formed. Therefore there is nothing other that can give counsel in such a case but the gods. They know all things and give signs to whomever they please by sacrifices, omens, voices and dreams'.[3] Since his time, bureaucracy has replaced the gods. Clausewitz, moreover, focused essentially on the operational level during war and it is this plane, rather than that of strategy, to which the comments in this article pertain. Clausewitz assumed that commanders should usually possess accurate intelligence about strategic issues, especially before war broke out. This, incidentally, explains why Clausewitz thought that strategic surprise was a good idea in theory but extremely difficult to execute in practice. 'The movement of the enemy's columns into battle can be ascertained only by actual observation . . . but the direction from which he threatens our country will usually be announced in the press before a single shot is fired.' 'But uncertainty decreases the greater the distance between strategy and tactics; and it practically disappears in that area of strategy that borders on the political.'[4] Similarly, whereas Clausewitz concluded that boldness was fundamental to operations, he specifically denied that it had the same place in strategy.

Clausewitz's views on operational intelligence were not simple, nor were they absolutely accurate, as shown by the experience of at least one contemporary. Devoting much personal attention to the task, the Duke of Wellington demonstrated that it was possible to collect useful tactical and operational intelligence on the enemy. A small but notable minority of British and American generals of that era adopted a similar view. From an examination of the same base of data as Clausewitz, the Baron de Jomini drew different conclusions about operational intelligence, and, over 2,000 years earlier, the influential Chinese strategist Sun Tzu praised – in fact, exaggerated – the military value of intelligence.[5] None the less, Clausewitz offered a powerful generalization about the effect of operational intelligence before the modern era, and these views were central to his theory. To Clausewitz, the lack of sure, current and effective intelligence causes uncertainty to dominate the field of battle, the psychology of command and the art of war. This is a major factor underlying the 'friction' which distorts all attempts to act on intentions in the operational sphere. So deeply do these ideas on ignorance and uncertainty shape his logic and his theory of command and war that amendments to them automatically force one to reconsider Clausewitz's arguments from the start. In fact, if these variables were to be assigned new values, then Clausewitz's equations would produce new solutions. This is true of Clausewitz's view that deliberate deception is almost always useless in war while intentional strategic surprise can rarely be achieved;[6] true, by extension, of even more fundamental matters, such as Clausewitz's belief that uncertainty and friction together prevent commanders from achieving precisely intended results precisely *as* intended; and true of his emphasis on intuition and determination in thought, and on boldness in operations, his theories of understanding, command and action.

According to Clausewitz, few men function well amidst the uncertainty and ignorance, the friction and danger, of the battlefield. That environment can be mastered only by the 'military genius', a man with 'a very highly developed mental aptitude for a particular occupation', war. This figure combines the faculties which 'taken together, constitute the essence of military genius. We have said in combination, since it is precisely the essence of military genius that it does not consist in a single appropriate gift – courage, for example – while other qualities of mind or temperament are wanting or not suited to war'.[7] Gifts of character, thought and action fuse into one to form the military genius. While pedantry – Clausewitz's favourite word of denigration – killed command, other intellectual powers gave it life. War 'cannot be waged except by men of outstanding intellect',[8] possessing 'a special type of mind . . . a strong rather than a brilliant one',[9] with 'a sensitive and discriminating judgement . . . a skilled intelligence to scent out the truth'[10] and presence of mind, the ability to respond quickly and competently to the unexpected.

When assessing the personal characteristics of commanders, Clausewitz used the language of ideal types and the logic of the golden mean, recognizing that either the lack or the predominance of any one characteristic could be costly and that desirable characteristics were linked on a spectrum to undesirable ones. Thus, carried too far, determination could 'degenerate into obstinacy'.[11] Moreover, he was never dogmatic: to every 'rule' or 'principle' he describes, he immediately notes the exception or the limitation. None the less, two faculties in particular were the bedrock of military genius. *Strength of character*, the 'ability to keep one's head at times of exceptional stress and violent emotion', allowed the reason of the commander to dominate his passions without destroying their drive.[12] Equally fundamental was *determination* – a willingness to 'stand like a rock',[13] to act on belief despite uncertainty, to hold to a consistent course of action amid confusion. Clausewitz commended the consistent pursuit even of an inferior course of action. He understood the old principle: order, counterorder, brings disorder. 'The role of determination is to limit the agonies of doubt and the perils of hesitation when the motives for action are inadequate . . . when a man has adequate grounds for action – whether subjective or objective, valid or false – he cannot properly be called "determined". This would amount to putting oneself in his position and weighing the scale with a doubt that he never felt.'[14] Determination 'is engendered only by a mental act; the mind tells man that boldness is required, and thus gives direction to his will. This particular cast of mind, which employs the fear of wavering and hesitating to suppress all other fears, is the force that makes strong men determined.'[15] Determination (in Clausewitz's sense) alone could prevent action from being paralysed by uncertainty and the delays and hesitation caused by thought.

The mind of the military genius combined reason and intuition – 'an intellect that, even in the darkest hour, retains some glimmerings of the inner light which leads to truth . . . the quick recognition of a truth that the mind would ordinarily miss or would perceive only after long study and reflection'.[16] It combined thought and temperament into one whole greater than its parts. 'Knowledge must be so absorbed into the mind that it almost ceases to exist in a separate, objective way . . . Continual change and the need to respond to it compels the commander to carry the whole intellectual apparatus of his knowledge within him . . . By total assimiliation with his mind and life, the commander's knowledge must be transformed into a genuine capacity.'[17] Clausewitz's statement that 'all great commanders have acted on instinct'[18] refers to this union of thought and temperament, which might be termed informed intuition. This faculty of *coup d'oeil* alone can master uncertainty and ignorance, and understand the battlefield.

> Circumstances vary so enormously in war, and are so indefinable, that a vast array of factors has to be appreciated – mostly in the light of probabilities alone. The man responsible for evaluating the whole must bring to his task the quality of intuition that perceives the truth at every point. Otherwise a chaos of opinions and considerations would arise, and fatally entangle judgement. Bonaparte rightly said in this connection that many of the decisions faced by the commander-in-chief resemble mathematical problems worthy of the gifts of a Newton or an Euler.[19]
>
> In the dreadful presence of suffering and danger, emotion can easily overwhelm intellectual conviction, and in this psychological fog it is so hard to form clear and complete insights that changes of view become more understandable and excusable. Action can never be based on anything firmer than instinct, a sensing of the truth. Nowhere, in consequence, are differences of opinion as acute as in war, and fresh opinions never cease to batter at one's convictions. No degree of calm can provide enough protection: new impressions are too powerful, too vivid, and always assault the emotions as well as the intellect.
>
> Only those general principles and attitudes that result from a clear and deep understanding can provide a comprehensive guide to action. It is to these that opinions on specific problems should be anchored. The difficulty is to hold fast on to these results of contemplation in the torrent of events and new opinions. Often

> there is a gap between principles and actual events that cannot always be bridged by a succession of logical deductions. Then a measure of self-confidence is needed, and a degree of skepticism is also salutary. Frequently nothing short of an *imperative principle* will suffice, which is not part of the immediate thought-processes, but dominates it: that principle is in all doubtful cases to stick to one's first opinion and to refuse to change unless forced to do so by a clear conviction. A strong faith in the overriding truth of tested principles is needed; the vividness of transient impressions must not make us forget that such truth as they contain is of a lesser stamp.[20]

INTELLIGENCE AND RISK-TAKING

This is the background to Clausewitz's art of war, the means by which the intelligent commander without intelligence should function in conditions of uncertainty. Boldness is central to this theory of operations.

> . . . In what field of human activity is boldness more at home than in war? . . . boldness in war has its own prerogatives. It must be granted a certain power over and above successful calculations involving space, time, and magnitude of forces, for wherever it is superior, it will take advantage of its opponent's weakness. In other words, it is a genuinely creative force.

To Clausewitz, 'a distinguished commander without boldness is unthinkable. No man who is not born bold can play such a role, and therefore we consider this quality the first prerequisite of the great military leader'.[21]

Daring was valuable because it made commanders act and because, *when they did act on its dictates, it multiplied all the conditions of* uncertainty which frightened their opponents. In fact, a bold commander tended to create timid enemies – his own confidence thrusting the burden of uncertainty onto the foe. While, in Clausewitz's view, a general might never be able to increase his own certainty, his boldness would *heighten the enemy's uncertainty and thus indirectly compensate* for his own limited intelligence. These observations remain true for all wars. Two hostile commanders stand at the opposite ends not only of the field of battle but also of a balance of uncertainty and an equilibrium of intelligence. This relationship is relative – by increasing the enemy's uncertainty, you decrease your own – and complex. Even when the quality of intelligence is better than assumed by Clausewitz, pursuit of an unexpected and high-risk strategy can still upset the balance of uncertainty; indeed, the subjective certainty engendered in a com mander's mind by better intelligence may actually increase his opponent's chances to achieve success through an unexpected strategy. Daring also maximized one's chances to surprise the enemy in general and even to surprise him in a specific way through deliberate intent. Clausewitz held that this was impossible to do on the strategic or strategic/operational levels but could be achieved at the operational or operational/tactical levels through a combination of boldness, determination, creative imagination and intellect. 'Only the commander who imposes his will can take the enemy by surprise; and in order to impose his will he must act correctly.'[22] Today, of course, the reverse is true – given such reliable and widespread devices as radar, complete surprise is almost impossible to achieve at the operational level and almost impossible to avoid at the strategic level.[23]

Scarcity multiplied the value of boldness. To Clausewitz, audacity is less dangerous in commanders than caution and less frequently found. 'Given the same amount of intelligence [in a commander], timidity will do a thousand times more damage in war than audacity'; 'a general cannot be too bold in his plans, provided that he is in full command of his senses, and only sets himself aims that he himself is convinced he can achieve'; 'We always have the choice between the most audacious and the most careful solution. Some people think that the art of war always advises the latter. That assumption is false. If the theory does advise anything, it is the nature of war to advise the most decisive, that is, the most audacious'; 'boldness grows less common in the higher ranks'; 'no case is more common than that of the officer whose energy declines as he rises in rank and fills positions that are beyond his abilities.'[24] In Clausewitz's equation, uncertainty and its psychological consequences reduce the frequency of boldness among commanders and increase the value of this scarce commodity on the field of battle. If all commanders were aggressive risk-takers, this characteristic would define the military art but it would no longer offer a comparative advantage to any general. When most commanders are cautious, then audacity is at a premium. In a world of risk-takers, the deliberately cautious man is king.

Yet boldness in itself is merely a matter of animal spirit and is inferior to temperament guided by reason. The few generals who combine these characteristics are the greatest of military geniuses.

> Boldness governed by superior intellect is the mark of the hero. This kind of boldness does not consist in defying the natural order of things and in crudely offending the laws of probability; it is rather a matter of energetically supporting that higher form of analysis by which genius arrives at a decision; rapid, only partly

> conscious weighing of the possibilities. Boldness can lend wings to intellect and insight; the stronger the wings, then, the greater the heights, the wider the view, and the better the results; though a greater prize, of course, involves greater risks.[25]

In the most desperate of hours, when material strength cannot prevail, the day may yet be saved by a commander who unifies 'boldness and cunning'[26] (not intellect but 'secret purpose', a combination of intuition and practical knowledge).

Clausewitz also noted a third combination of temperament and reason, one pointing away from the others and also from all of his own generalizations regarding the psychology of command. Caution was generally produced not by choice but by character, by an instinctive reaction to uncertainty. Worst-case analysis, he argued, was the natural response to the unknown. 'Men are always more inclined to pitch their estimate of the enemy's intentions too high than too low, such is human nature.'[27] Sometimes, however, caution was the deliberate decision of a superior or cunning intellect. This unusual combination of characteristics was the most balanced, the most powerful of them all.

> Whenever boldness confronts timidity, it is likely to be the winner, because timidity in itself implies a loss of equilibrium. Boldness will be at a disadvantage only in an encounter with deliberate caution, which may be considered bold in its own right, and is certainly just as powerful and effective; but such cases are rare. Timidity is the root of prudence in the majority of men.[28]

Certain dimensions of Clausewitz's arguments concerning 'boldness' and 'caution' merit careful attention. His main tool for analysis and synthesis was the ideal type method. He distilled complex issues into their basic components and developed his concepts by posing simple and often extreme or contradictory statements against each other. Hence, if one takes any statement out of context, one distorts his entire argument. The context for his discussion of 'boldness' and 'caution' is Book Three of *On War*, which defines the general phenomena that shape strategy as a whole. Here Clausewitz uses a conceptual model of the relationship between moral and mental qualities, and explains how all of these matters affect generalship as a whole. Not surprisingly, given this context, he treats commanders as if they possess one and only one psychological characteristic. They are 'all bold' rather than a mixture of characteristics, such as 'mostly bold with a shade of caution'. He also discusses the military effect of these psychological traits in the abstract, that is, how the characteristics of 'boldness' or 'caution' are affected by and affect the condition of 'uncertainty'. His generals are avatars of an ideal type rather than human beings. Clausewitz does not discuss in precise historical terms how the relative 'caution' or 'boldness' of one commander fighting another works in specific circumstances. Indeed, no doubt he would have regarded this point as pedantic, since he considered these characteristics to be absolute rather than relative; that is, one is either bold or not, and when comparing two audacious commanders the point is not that one is more daring than the other, but that both are daring. Given his concerns and his level of analysis, it was reasonable for Clausewitz to adopt such a means of assessment, but commentators must take care in transferring his generalizations about psychological characteristics to their own assessments of specific commanders. What is true of ideal types need not be true of individuals, especially not when, as will be shown below, 'uncertainty' – the central component in his theory – is open to revision.

Nor is it ever easy to define psychological characteristics such as 'boldness'. Are they determined by a general's behaviour – his usual or universal practices when selecting options – or by his essential and innate temperament? In any case, few commanders always behave in what is commonly regarded as an 'audacious' or 'cautious' manner. In the Second World War, for example, Rommel was daring in the desert but not at Normandy. Montgomery usually adopted a deliberately cautious approach, but he veered occasionally toward both daring and timidity. The behavior of most commanders fluctuates within a given range on the spectrum between these traits.

Above all, confusion arises from terminology. In common usage, 'boldness' is defined by such words as 'aggression' and 'high risk' and is identified with field commanders like Rommel. This is not Clausewitz's usage – or, more precisely, not his only usage. He defines daring by reference not to a few specific sorts of behavior but to the ability to act in an unexpected fashion. Under his usage, if everyone were to follow a mindless cult of the attack, then the risk-taker would be reckless, while the truly bold commander might be the one who turns to a carefully calculated defensive battle. This is especially true because he defines defense rather than any form of attack as being the strongest from of war. Clausewitz, of course, did not regard such circumstances as the general rule. In practice he usually did associate boldness on the battlefield with aggressive and high-risk actions. None the less, paradoxical as it might seem, in the Clausewitzian sense 'boldness' could sometimes be defined by such words as 'caution' and 'calculation', if used in a creative or unexpected way. To Clausewitz, Fabius Maximus or Pericles would be 'audacious' – indeed, Montgomery at Alam el-

Halfa might have seemed a bolder commander than Rommel, by daring to use the tactics of 1918 against those of 1940, in an unexpected fashion and with unexpected success.

For example, Clausewitz defines deliberate caution as being 'bold in its own right', presumably because, in his theory, this stems from the same combination of thought and temperament which produces 'boldness governed by superior intellect'. In another context, Clausewitz emphasizes that 'when a man has adequate grounds for action – whether subjectively or objectively, valid or false – he cannot properly be called "determined" '.[29] The same logic can be applied to 'caution' or 'daring': the more adequate the grounds for action, or the greater the degree of certainty, the less significant the effect of psychological characteristics on decisions. To Clausewitz, daring is essential in circumstances of uncertainty, where it substitutes for lack of strength or intelligence. If, however, one possesses accurate and usable intelligence on the enemy as well as greater strength, then what seems to be a bold action by common usage would simply be an act of calculation in Clausewitz's teminology – a calculated risk. In such circumstances calculation can replace boldness. If perfect intelligence could be acquired on the operational plane, daring would be no more relevant to a commander than it is to a chess player choosing whether or not to sacrifice a piece. Little daring is required to bet on a sure thing, little caution to avoid certain destruction. Intelligence can turn caution into daring, and daring into calculation. One must not confuse a calculated action with a bold one, nor take aggression for audacity.

Hereafter we will refer to the common usage of 'boldness' with the phrase 'aggressive risk-taking', as distinct from 'boldness (in Clausewitz's sense)'. At different times, the analysis will focus on two distinct issues – how intelligence affects (a) the range of behavior of specific commanders on the spectrum between 'caution' and 'boldness' and (b) the relative value on the battlefield of these psychological characteristics as ideal types, in the age of accurate and real time intelligence.

It is on the foundation of uncertainty that Clausewitz formulated his definition of the ideal general's psychological characteristics and all his normative rules for command – especially the '*imperative principle*' ('in all doubtful cases to stick to one's first opinion and refuse to change unless forced by a clear conviction') and the value of boldness. Only the 'harmonious combination of elements', which make up 'the essence of military genius – in particular, daring, determination and informed intuition – can ward off the psychological threat of uncertainty. 'With uncertainty in one scale, courage and self-confidence must be thrown into the other to correct the balance.'[30]

During this century, however, changes in the status of intelligence have at times eroded – though not ended – the reign of uncertainty over the field of battle. Ultra and uncertainty may represent the opposite poles of a commander's understanding in war. This is not to say that Ultra represents perfect knowledge, but rather the closest approximation to perfect and usable knowledge that commanders can expect to receive. How has this affected the validity of the Clausewitzian paradigm? How has increased knowledge affected the psychology of command, the art of war, the qualities required for military genius and the relative value of caution and boldness? If intelligence provides 'adequate grounds' for action, what significance do daring and determination (in Clausewitz's sense) retain? Any answers to these questions must take three general points into account.

First, intelligence does not produce command decisions, a commander does. His personality, attitude toward risk, and military education are therefore the most important factors in the use of intelligence and the practice of operations.

Second, the greater the influence of Clausewitz's analysis of the psychology of war and the more widely accepted his prescriptions for handling it, the less their accuracy. Clausewitz defined the means to manage uncertainty without reducing its scale, particularly through determination in the mind of the commander and boldness in his deeds. In the century and a quarter following his death his theory had mixed fortunes, but his prescriptions regarding these specific characteristics were influential, largely because they fitted with those of other authorities and with the lessons which officers intuitively derived from the Napoleonic and Nelsonic experience.[31] Many armies, pre-eminently those of Germany, Japan and France before 1914, did follow his prescriptions, in a vulgarized fashion; simultaneously, the Nelsonic tradition affected the British, German, Japanese and American navies in a similar way.

John Keegan has recently argued that Clausewitz's views are irrelevant to historians because they have never influenced soldiers.[32] One might as well argue that Newton has never influenced a gunner. Nor can Keegan's view survive contact with primary documents. The writings of several American generals of the Second World War, for example, are permeated with both pure and vulgarized forms of Clausewitz. Consider George S. Patton's lecture of 1926 on 'victory':

> War is an art and as such is not susceptible of explanation by fixed formulae . . . War (is) . . . violent simplicity in execution (and) pale and uninspired on paper . . . Insist on: Maximum distances;

> BOLDNESS; Prompt and accurate reports. Speed is more important than the lives of the (men at the) point. THE ENEMY IS AS IGNORANT AS YOU . . . You must not be delayed by blufs [*sic*]. Bluf [*sic*] yourself . . . BE BOLD . . . YOU ARE NOT BEATEN UNTIL YOU ADMIT IT, Hence DON'T . . . The 'Fog of war' works both ways. The Enemy is as much in the dark as you are. BE BOLD!!! [*sic*].

Similarly, in 1943 Patton wrote:' Never take counsel of your fears. The enemy is more worried than you are. Numerical superiority, while useful, is not vital to successful offensive action. The fact that you are attacking induces the enemy to believe that you are stronger than he is . . . IN CASE OF DOUBT, ATTACK.' Even Dwight Eisenhower insisted that instructors should

> treat war as the drama that it is rather than constantly reducing it to a science of marching tables and tonnage calculations. I do not decry the necessity for the scientific end of the education, I merely think that too many officers develop their thinking more and more along lines of mathematical calculations rather than realizing that calculations always go wrong.
>
> Many generals constantly think of battle in terms of, first, concentration, supply, maintenance, replacement, and, second, after all the above is arranged, a conservative advance. This type of person is necessary because he prevents one from courting disaster. But occasions arise when one has to remember that under particular conditions, boldness is ten times as important as numbers.[33]

None of this proves a direct connection between Clausewitz's ideas and these commanders' actions, but it does show that they explained their views in Clausewitzian language. Until Keegan can come to terms with the evidence, his assertions are useless.[34]

Generals, of course, were not Clausewitzians in a conscious sense. In taking fractions of his argument out of context and placing them in the context of the Napoleonic tradition, soldiers turned his precepts on their heads – thus, Clausewitz's insistence on the importance of determination was used to defend obstinacy, his arguments about boldness influenced a mindless cult of the offensive, his idea of the military genius was used to justify the idea of the commander as a great man – or dictator. The point is simply the effect of this vulgarized variant of Clausewitz. While boldness (in Clausewitz's sense) remained as rare in this century as ever, aggressive risk-taking did become more common among commanders than usual. In principle, as aggressive and high-risk behavior on the battlefield becomes more common, it changes the context which defines its own value: the more numerous the aggressive risk-takers, the less the advantage offered by that characteristic. The same is true of determination (in Clausewitz's sense). Increasingly, as aggressive risk-takers fight each other, both will lose the operational benefits stemming solely from this characteristic: indeed, the least aggressive of them might have the advantage. The more common aggressive risk-taking becomes, moreover, the more other (and temperamentally cautious) commanders will be forced to find a counter to it, and the more they will succeed in doing so. For commanders on the battlefield, psychological characteristics do not have an absolute value but one defined relative to those of their adversaries. With an increase in aggressive risk-taking, other commanders are given an incentive – an imperative – to anticipate then to minimize its potential advantages. Thus, one might conclude, as a direct result of his observation, empirically based and tolerably accurate, that aggressive risk-taking tended to become less effective than Clausewitz predicted and caution, though not timidity, more so.

Third, many commentators, merging Clausewitz, the cavalry spirit and the notion of 'Great Captains', have treated aspects of war such as boldness of command and skill in maneuver as though these were universally the most – perhaps, the only – fundamental prerequisites for victory. The following points apply not to Clausewitz but to views common among modern commentators. This view is most commonly attached to the influential ideas of Basil Liddell Hart although many other military commentators have expressed views. George S. Patton also spoke of the 'Great Captains' and 'the immortal line of might warriors'.[35] Exponents of such ideas have tended to view war in romantic terms (as a form of art) and to adopt such beliefs as the idea that mobile operations must represent a higher form of the operational art than set-piece battles. Although Clausewitz himself despised this approach, it is particularly pronounced among members of the romantic school of military history, with their adulation of the romantic school of military commanders, especially in its German form. They involve several fallacies. These views fall prey to the idea of the 'master cause': in effect, they treat just a few of the many factors involved in a process as being the only necessary – and therefore sufficient – causes for it. As a result, certain factors are overemphasized while others, perhaps equally important, are treated as passive conditions and therefore forgotten.[36] The views in question, moreover, are ahistorical. The role of each individual aspect of command and military prowess varies with time and place. There is no universal value for any one of these characteristics nor any constant basket of weights for the whole: the importance of all these matters varies from one period to another. *Ipso facto*, this is also true of the characteristics of any individual commander. Furthermore, one common means used to determine the nature and the value of these characteristics – the comparison of commanders across the ages and the pursuit of the 'Great Captain' – is useful as a heuristic device but fatally misleading when applied as a precise and meaningful tool of analysis. 'Great Captains' are not people, they are a Platonic ideal – a way of gauging the quality of individual commanders. This ideal, in turn, rests on implicit assumptions about the universal value of specific military characteristics; that is, intellectual presuppositions are smuggled into an analysis and then uncovered as empirically demonstrated conclusions. Certain military qualities are fetishized, while others are ignored, without any rigorous justification being offered for this selection of values, then certain commanders are treated as archetypes for these allegedly universal qualities, as if Hannibal's characteristics would have automatically made him master of the western front. While a historical record demonstrates the usual (though not universal) inferiority of a Blücher to a Bonaparte, how can one meaningfully compare two contemporaries who never fought each other, let alone commanders from different epochs; and from these hypothetical comparisons, how can one derive lessons regarding the universal value of any single characteristic of command or of war?

These issues of theory can be illuminated through a study of the war fought by Douglas MacArthur and his lieutenants against powerful but ignorant Japanese enemies, who, instead of using intelligence to control uncertainty, relied on a vulgarized version of the Clausewitzian art of war, combined with selected traditions from their own warfare. These conclusions can be compared with Allied experiences in Europe between 1942–45 against a German enemy with similar characteristics. This intelligence record will provide a mirror for generals and generalship: by comparing what a given officer believed with his actions one can reconstruct his calculus of military actions and his nature as a commander.

ULTRA IN THE PACIFIC

Seven months in the Pacific produced two of the most celebrated events in the history of intelligence. While ample attention has been paid to the role of codebreaking in the United States' disaster at Pearl Harbor and its triumph at Midway, the studies of Ultra (or 'special intelligence', as American officers more commonly called it) in any other part of the Pacific war have been limited in quantity and quality. A bridge has now been thrown across this gap in scholarship. The primary strength of *MacArthur's Ultra, Codebreaking and the War Against Japan, 1942–1945*[37] is Edward Drea's analysis of how one source of intelligence worked how it fitted in with the rest, how the material which it produced was assessed and used by commanders in a single theater and how all of this affected the course of one campaign. The evidence on these issues is better than usual in the study of military intelligence and Edward Drea has mined it with energy and craft. Drea, a master of the American records on MacArthur's campaigns, has also utilized the surviving Japanese documents with unprecedented thoroughness. *MacArthur's Ultra* possesses greater analytical sophistication and covers more chronological terrain than the best previous accounts of intelligence during the Pacific war (and good ones, the works of Gordon W. Prange, *et al.*).[38] It is entirely superior in analysis, synthesis and research to lesser but still useful studies, such as those by John Costello, Edwin Layton, Ronald Spector and Ronald Lewin. Drea's sober and multi-causal account is far from the speculation, sensationalism and exaggeration of the role of intelligence which mar so many studies such as those by Ladislas Farago and James Rusbridger, and even those of Lewin and Costello.[39] Drea does not stretch evidence, misquote it, invent it or ignore it, nor does he ever forget that intelligence alone can never win a war. On the other hand, he demonstrates that Ultra was the decisive source of intelligence in the southwest Pacific, one which fundamentally shaped the course of the Pacific war.

MacArthur's Ultra is essential on two counts. It is the first serious study of signals intelligence in the southwest Pacific theater and it is the best single study of that campaign as a whole. While Edward Drea is far from ignorant of the practice of intelligence, he is not a specialist in its study. Thus, he is immune to the ills specialists suffer, particularly the tendency to exaggerate the significance of intelligence. Drea's analysis of the technical aspects of the topic – the communication and cryptographic systems of the Imperial Japanese Army and Navy (IJA and IJN), the techniques by which Allied cryptanalysts attacked these codes and the process by which the product was distributed, analysed and utilized – is useful to the expert while comprehensible to the common reader. His account of codebreaking organizations and the role of data-processing machines as a cryptanalytical tool in the South West Pacific Area Command (SWPAC) will particularly interest students of cryptology. Nor does Drea overrate the role of codebreaking. He places it in the context of all sources of information and he pays unusual attention to often overlooked elements of signals intelligence, such as traffic analysis. No other book about the Second World War explains with more authority exactly what information was available to any Allied commander and his staff on a daily basis and how this affected their decisions. Few match the rigor with which Drea compares the causal role of Ultra to that of its fellow factors – operational, logistical, doctrinal, topographical, personal – in the shaping of events. This work ranks alongside those of Ralph Bennett among the best, not to mention the few, scholarly studies of signals intelligence during the Second World War.[40]

Drea takes as his text the dictum of Bennett and F.H. Hinsley that there was not one Ultra but several: that the role and importance of special intelligence varied with time and theater. Certainly there were fundamental differences between the Ultras of the Pacific and Europe, for technical, tactical, organizational and political reasons. In Europe, Ultra emanated from a centralized bureau controlled by one country, which co-ordinated the attack on every code of two others. Codebreaking against Japan was produced by several agencies of three allies (Australia, Great Britain and the United States) working in a decentralized fashion. The USN and the American Army ran separate organizations, each divided between autonomous bureaus in Washington and the Pacific, while the cryptanalysts in Delhi had greater independence than any other sub-section of British codebreaking. This frustrated those at the time who held that only a centralized system could foster efficient cryptanalysis and that frustration has shaped the historiography of the period. It has led, in particular, to the claim that because SWPAC sabotaged centralization, it must have abused Ultra.[41] This may be true but it has yet to be proven. Certainly the loose system of the Pacific met the needs of the USN in 1942, while Drea argues that it always worked for SWPAC. However effective in technical terms, this system had obvious political consequences: it left ample room for bureaucratic buccaneering. Nothing could have better suited MacArthur, a commander of messianic self-confidence, a well-honed public relations machine with remarkable political power, a desire for total sway over his own sphere and the determination to control Allied strategy against Japan. He had unique control over codebreaking in his theater for 'Central Bureau', MacArthur's signals intelligence agency, was never subsumed in the integrated Anglo-American cryptanalytical system. This remained his private intelligence bureau, producing MacArthur's Ultra for MacArthur's aims – to crush the foe and influence his friends.

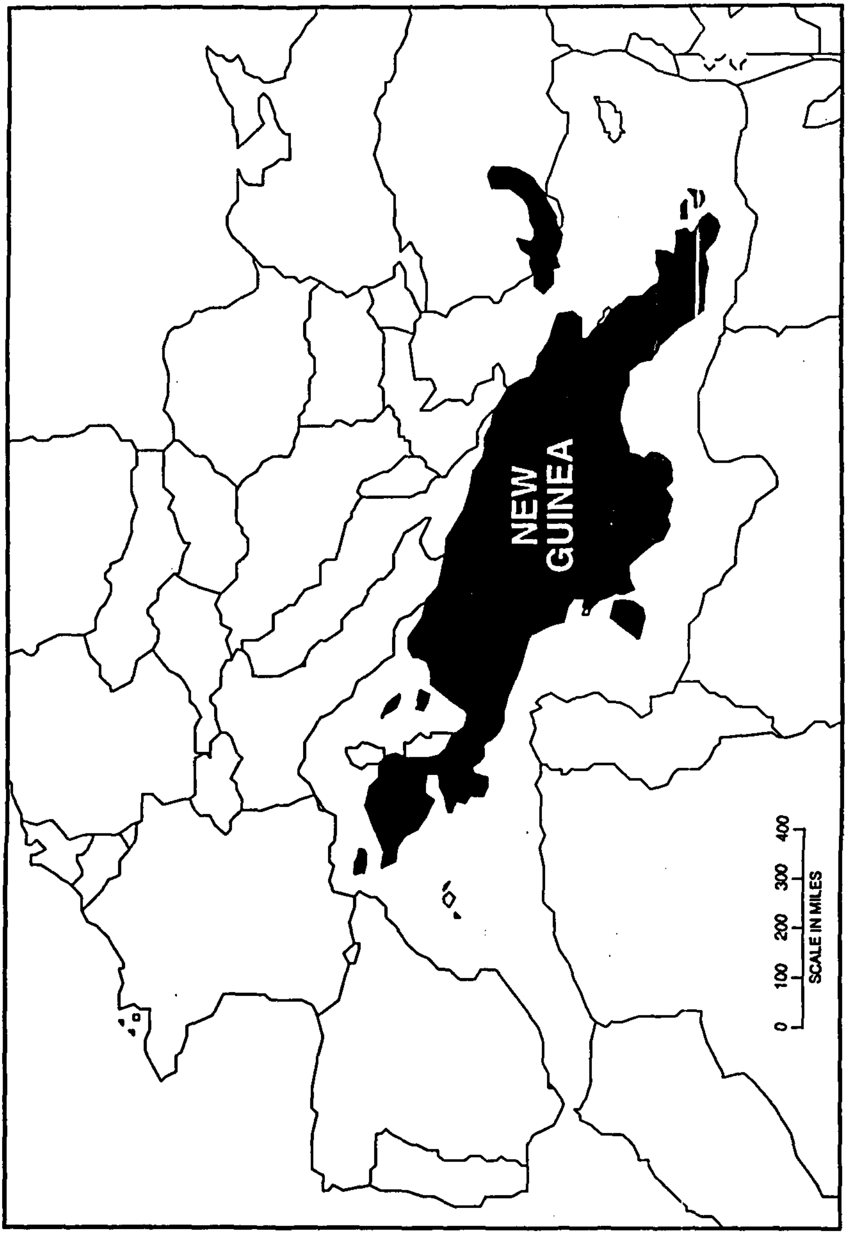
Drea, by implication, leaves his readers with the impression that Ultra in Europe grew technically more successful and militarily more useful in a steady, regular and virtually unbroken fashion as time went by. While this argument is tolerably accurate as a broad generalization for the whole period 1939–45, it is not true of specific matters during smaller periods of time, such as the campaigns in the Atlantic and in Africa during 1940–43. The history of Ultra in these two theaters and in the southwest Pacific is very similar, replete with sudden and complete cryptanalytical successes or defeats and reversals of cryptological fortune – situations in which a total inability to penetrate one of the enemy's cryptographic systems coincided with total access to others. Moreover, in these campaigns technical achievements in cryptanalysis and success on the field of battle were not linked in a simple linear pattern. Arguably, in the Mediterranean campaign Ultra could have been most useful when it was technically most primitive rather than most mature, because of the prevailing strategic and operational conditions. When it was most primitive, force-to-space ratios were low, as were both sides' strengths; hence, victories with decisive consequences were feasible. By the time that Ultra matured, both sides were locked in a prolonged struggle of attrition.[42] The differences between the history of Ultra in Europe and the Pacific are not as stark and simple as Drea suggests. They also exist in areas which are not discussed in his formal and explicit analyses. In his formal explanations of the causes for events, Drea appears to explain the differences between the effect of Ultra in Europe and the Pacific primarily in terms of the technical success of cryptanalytical bureaus and the personal predilections of commanders. He does not always explicitly incorporate two other factors: Ultra's limitations as a source of intelligence – what material it could and could not provide – and the issue of how far such material could and could not be used in distinct operational and strategic circumstances. On the other hand, these matters are integrated implicitly into his narrative, and effectively so.

The history of Ultra in Europe and the Pacific also differed. In Europe the sequence of events regarding Allied success against German cryptographic systems was broadly similar, although the date at which the process began varied significantly with the German service in question. First came a period of limited and often fragmentary access to specific Enigma keys, coupled with major problems in assessing such material. Then followed a cryptanalytical breakthrough, which produced high-level material on a continual and current basis; finally, analysts enjoyed a long period of mature exploitation, during which they determined the main lines of the enemy's organization, drew powerful conclusions about its operational intentions and precise ones about its capabilities, and were almost always accurate about such fundamental matters as the enemy's order of battle.

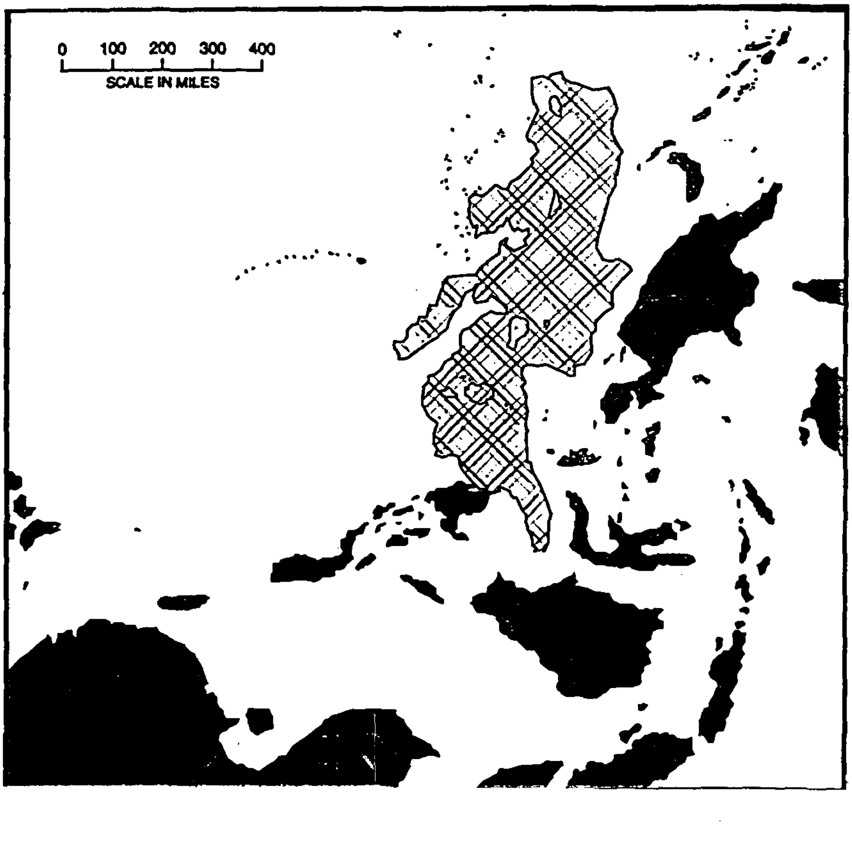
Little of this was true of the Pacific, although that theater was at first glance ideally suited for signals intelligence. Forces were scattered over millions of square miles, often lacking direct links by cable; hence radio dominated communications. Prisoners of war and agents were less effective sources of intelligence than usual while captured documents, photo reconnaissance and signals intelligence were far more important than usual. In the wireless war the Allies had two additional advantages: the IJA and IJN lacked sophisticated signals intelligence services and placed immense faith in the formal security of their cryptographic systems. The operational circumstances of the Pacific enhanced these opportunities. (See Maps 1 and 2.) Force-to-space ratios were extraordinarily low. Most elements of both sides were rarely in direct contact with the enemy; consequently the disposition, strength and intentions of such forces were masked and difficult to calculate. Seldom has possession of the initiative carried such consequences. It was unusually difficult to recover one's balance if struck by an unexpected blow – weeks might be required to redeploy naval or air forces from one base to another, months to build the infrastructure needed to maintain large forces in a new area or to move soldiers by sea or land. The war in New Guinea, in particular, was a sideshow in which neither side could easily replace its losses. The destruction of 20,000 men or 200 airplanes or the capture of one supply base could transform a campaign in a territory with the geographical space of the Mediterranean theater. Operations in New Guinea, for example, turned on the annihilation of forward Japanese airbases in August 1943 and on the battles of the Bismarck Sea and Hollandia, and in all of these absolute Japanese material losses were quite small. The ability to concentrate strength against enemy weakness, to catch the enemy by surprise and to profit from a knowledge of enemy intentions was unusually great; this was doubly true for that most risky and complex of military operations, amphibious assaults. Such circumstances generally provide the greatest opportunities for success through signals intelligence.

Events did not work quite as might have been anticipated, because of the IJA's organization and its cryptographic systems. The IJA and IJN relied on superenciphered codebooks, which were slower to use than the machine cipher systems of the era. They were more labour-intensive and unable to carry as heavy a load of traffic before coming cryptologically vulnerable, while the distribution of new books and tables was dogged by difficulties; nevertheless, they were a perfectly good means of defense if used with parsimony and skill. In practice, they were as secure as the technically more sophisticated Enigma and Purple systems; at least changes in superencipherment tables sometimes denied

MAP 1
THE NEW GUINEA AND MEDITERRANEAN THEATERS COMPARED



MAP 2
THE MEDITERRANEAN AND WESTERN PACIFIC THEATERS COMPARED



Allied cryptanalysts complete access to Japanese codes. Moreover, these systems often were used with skill, if not always so. From early 1942 the IJN's most important radio signals were more frequently read by the enemy than not, although access might be lost for several weeks at a time when codebooks or superencipherment tables were changed. By December 1941, however, none of the IJA's systems had been penetrated, and thereafter its security procedures were sound. They were always better than the practices of the armies in Germany, Italy and the Soviet Union and, until 1944, at least equal to those of the American or British Commonwealth forces. There was one solitary if crucial exception to that rule, the IJA's 'water transport' system, used for army traffic about troop and logistical shipments by sea. This was created in December 1942 and solved by Allied cryptanalysts four months later.[43]

Thus the Allies had more or less complete mastery over all codes regarding Japanese maritime matters. That was the fundamental priority in this genuinely combined services war, where the struggles on land, sea and air were far less self-contained than in Europe. However, the IJA's cryptographic systems for use within divisions (and sometimes between garrisons and their immediate superiors) were the most secure of any army of the entire Second World War. They were never once solved. Admittedly, the elements within Japanese divisions needed relatively little intercommunication by radio, given their operational circumstances: had they been forced to signal to the degree required under European conditions, either Japanese command within divisions would have been strangled or the security of their codes would have collapsed.[44] The IJA had not found a general solution to the problem of middle-grade cryptographic systems and communications within divisions, merely an effective solution to its own particular difficulties in that sphere. Until January 1944, Western cryptanalysts also failed to penetrate the superenciphered codebooks used for communication between Japanese formations. Only when the Japanese 20th Division failed to destroy its codebooks during a withdrawal in New Guinea did the Americans finally break in, assisted by subsequent compromises of a similar nature.[45] As so often occurs, defeat on the field of battle crippled signals and cryptographic security, and compromises of security fathered operational catastrophes, in a vicious circle. None the less, for the first half of the Pacific war, more signals intelligence about Japanese land operations was acquired through cryptanalysis of naval than of army traffic, just as at the same time solutions of Luftwaffe traffic were the main source of Ultra on the German Army. Of all the Axis fighting services the IJA, along with the German U-boat command, possessed by far the hardest cryptographic shell. In both cases the nut cracked only because the Allies acquired copies of the system, unbeknown to the enemy. Hence, from the spring of 1942 Allied commanders in the southwest Pacific could profit from traffic analysis against IJN and IJA stations, along with mastery of the IJN's operational traffic, its 'water transport' systems and, after April 1943, of the IJA's 'water transport' systems. Until January 1944, however, rather late in the day, they had no access to the IJA's operational traffic. After that date, material of this sort flooded in.

Nor was that the only problem at hand. In both world wars, the primary task of signals intelligence was to master the enemy's order of battle, after which its capabilities were much easier to define and its intentions less difficult to determine. This task was most readily accomplished against an enemy with a predictable chain of command. It was unusually difficult against the IJA because the success of signals intelligence varies with the cultural, structural and doctrinal background of the target. The IJA did not organize on the conventional Western pattern. To a greater extent than any other army of the Second World War, Japanese commands were established in the form of ad hoc task forces, composed of a unique melange of units drawn from many different formations, which were often designated only by the name of their senior officer; elements of a single division were routinely dispersed hundreds of miles from the others. The location of a given German regiment normally indicated the presence of a specific division. The strength of such forces usually could be calculated with a tolerable degree of accuracy, and sometimes with extraordinary precision. This was not true of the IJA nor of many non-Western armies during this century, such as those of the Turks during the First World War and the Iranians during the Gulf War of the 1980s.[46] Structural links between the IJA's units and formations were loose, by Western standards: the identification of a given unit did not necessarily indicate anything about any formation, or vice versa. When reconstructing the IJA's order of battle, the basic building block was the battalion rather than the division: this rendered reconstruction more difficult by an entire order of magnitude. In 1937 the British Military Attaché in Tokyo, F.S.G. Piggott, who had controlled intelligence for the Second British Army during the last phase of the great war wrote that 'it was easier to draw up the German Order of Battle in France in 1918, than the Japanese Order of Battle in China in 1937'.[47] This difficulty persisted in various forms throughout the Pacific war. Until early 1944, the Allies had a surprisingly limited grasp of the enemy's order of battle both in theaters and garrisons. Then, through the break into IJA operational traffic and the difficult technical task of assessing 'address groups' (the elements within any cryptographic system which identify the transmitter and recipient of messages) Central Branch was able to determine the location, boundaries and titles of all IJA headquarters, and therefore the disposition and deployments of all its higher formations. Here, as ever, intelligence was entirely irrelevant to grand strategy, and secondary in the formulation of strategic objectives at the theater level – such issues are decided by the interests of states and the attitudes of statesmen. The material from Ultra, however, was sometimes fundamental to the execution of theater-level strategy. It frequently guided the selection between and the planning for major operations, and it was essential for the success of the most effective operational options available to SWPAC, outflanking or bypassing. At the same time, however, it was too imprecise to meet many critical operational needs, such as uncovering with certainty the strength and composition of specific commands.

Compared with the norm with signals intelligence, Ultra in the southwest Pacific was unusually effective in tracing enemy intentions but surprisingly unsuccessful in determining its capabilities – usually an easier task. Compared with its peak in Europe, Ultra in the Pacific never achieved complete mastery over the enemy's order of battle. After accounting for these differences, the two were roughly equal as sources of strategic intelligence and had A roughly comparable effect on strategy and the planning of major operations. The situation was more complex at lower levels of war. Compared with western Europe, Ultra in the southwest Pacific was generally a less thorough and accurate source of operational intelligence, but conditions on the battlefield allowed intelligence more easily to shape dramatic operations. Therefore, in the South-West Pacific, Ultra of technically lesser quality contributed to operational triumphs which matched anything in Europe. In the Pacific, Ultra's greatest value lay in addressing the issues of how to execute lines of strategy and where (or where not) to begin major operations. Once direct contact was established with the enemy, however, most information was acquired through tactical means in which the Allies had no particular advantages, which meant that the balance of intelligence between the IJA and the Americans moved toward equality.

These characteristics stemmed not merely from Japanese cryptographic and military-bureaucratic systems, but from the nature of Ultra and of intelligence assessment at SWPAC. Ultra was the most sophisticated source of intelligence during the Second World War, but it was never perfect and not always the best. Ultra took words straight from the enemy's mouth, but those words were not always straightforward.

As Ralph Bennett has written: 'Ultra always told the truth, but not always the whole truth.'[48] Ultra's output fell between two poles, one best described by words such as 'transparent' and 'self-evident' and the other by terms like 'opaque' and 'ambiguous'. The meaning of messages of the first type seemed to be intuitively clear compared with the framework of common sense, as when an enemy commander outlined his imminent intentions, or his quartermaster defined their order of battle, or a convoy signalled its location. In all such cases, the nature of the content was intuitively self-evident. Conversely, the content of traffic of the second type was, in itself, ambiguous – its meaning could be determined only through formal and frequently elaborate analysis. Fragmentary bits of intelligence about the enemy's dispositions and order of battle, for example, often provide value only through the derivation of inferences that can be acquired solely through the application of a reasoned, but seemingly arbitrary and arcane, process. The first sort of material is relatively easy to understand and trust and use; the second is more problematic on all of these counts. MacArthur's headquarters handled matter of the first kind very well but not the second. It could not master the higher level of analysis needed to exploit such material, work which Bletchley Park and Allied commands in Europe did so well. Hence, SWPAC's intelligence assessments were flawed and often dismally bad.

In this sphere MacArthur's headquarters was the least successful and among the least competent senior commands of the Western Allies. Here the critics of SWPAC's use of Ultra find much ammunition. Drea does not quite recognize the weakness of SWPAC's performance, but close reading of his narrative proves the point. He describes a constant pattern. SWPAC made effective use of unambiguous information, 'hot' and 'hard', about enemy intentions and capabilities, while its assessments of ambiguous material were mediocre to poor. In particular, SWPAC misconstrued the enemy's strength and dispositions in its planning before battle after battle. While such errors were most routine and extreme during the first year of the war, they occurred even to its end.

This stemmed primarily from the personality of MacArthur himself. He was not the type of officer to encourage the objective analysis of intelligence. He wished to see material which supported the aims he had already defined and proved they were good. Anything else he would ignore: nothing else could more easily be baked to suit his appetite than ambiguous material. MacArthur defined the nature of his meals through the selection of a cook who would always season his meals according to his master's taste. His intelligence chief, General Charles Willoughby, was almost always significantly wrong whenever his data were not self-evident and extraordinarily precise. Even then his estimates were usually mediocre, with a few brilliant exceptions derived from self-evident Ultra traffic, most notably before the attack on Hollandia in 1944. MacArthur once stated that there had been only three great military intelligence chiefs in history 'and mine is not one of them'.[49] This was an understatement: under MacArthur, Willoughby was arguably one of the three worst intelligence chiefs of the Second World War. It was also self-condemnation. Intelligence can only be as good as its master. Enlightened and open-minded generals will always gain more from it than will those who know what they want to be told and refuse to hear anything else. Some men use lamp-posts for illumination, some simply for support; and several subordinate headquarters under MacArthur's command, American or Australian, did assess Ultra on the IJA better than SWPAC.

MacArthur, Willoughby and other staff officers at SWPAC also failed to understand the Japanese way of war, the strategic calculus which determined what range of options Japanese commanders believed they had and how they would choose among them. Drea does not systematically assess American preconceptions about Japan and its army and their effect on American military operations (although few historians are better suited to this task), nor does he entirely share the influential views of John Dower on that topic.[50] Drea does, however, demonstrate that SWPAC's predictions of Japanese behavior rested on cultural and military ethnocentrism, ignorance of the enemy's doctrine and the belief that the enemy must act as Americans would in their place. Not surprisingly, right to the end of the war, Japanese decisions continually surprised SWPAC and many, although not all, of its subordinate commanders. The Australian General Thomas Blamey and the American General Walter Krueger grasped the IJA's style of operations quite well and that knowledge shaped their professional if unpublicized operations against it. Ultra, which sometimes saved MacArthur from the costs of such errors, was perhaps the only kind of intelligence capable of overcoming the combination of ethnocentrism and mediocre assessment at SWPAC as well as the unusual problems involved in determining the enemy's capabilities. Thus Ultra may have been even more crucial to MacArthur's success than Drea allows, yet he allows a great deal.

In the Pacific war, the period between August 1942 and September 1943 was one of equilibrium and attrition between forces roughly equal in quantity, quality and initiative. The build-up of Allied strength in New Guinea and the Solomon Islands turned the forward edge of

Japanese conquest into the central battlefield of the Pacific war. The operations in these areas cost Japan more than the United States. Japan was less able to replace losses, because of its casualties in aircrafts, pilots, transport vessels and soldiers substantially exceeded those of the Allies, who used their forces with greater economy and effect. In this campaign, the sources and targets of military power were armed men and the means to keep them armed. Both sides strove to maintain large forces in some of the least developed places between the poles, where virtually no logistical infrastructure or networks of physical communications existed. In such a war and such an environment, the United States held the strongest cards. Nowhere could it more thoroughly outmatch Japan than in creating an entirely new system of military logistics and in fighting a war of material attrition. Japan, conversely, had made precisely the errors defined by Clausewitz:

> Since the object of the attack is the possession of the enemy's territory, it follows that the advance will continue until the attacker's superiority is exhausted . . . What matters therefore is to detect the culminating point of victory with discriminate judgement.
>
> One defense is . . . not exactly like another, nor will defense always enjoy the same degree of superiority over attack. In particular this will be the case in a defense that follows directly the exhaustion of an offensive – a defense whose theater of operations is located at the apex of an offensive wedge thrust forward deep into hostile territory . . . a defense that is undertaken in the framework of an offensive is weakened in all its key elements . . .[51]

Instead, expanding until it could go no further, Japan reached its level of incompetence, where it fought with all its power and to the death. The battlefields of New Guinea and the Solomon Islands were simply too large and too far forward to suit its tiny logistical and transport capabilities. With every mile added to the distance between the battles and its main centers of power, Tokyo, Manila and Singapore, its logistical problems multiplied and its lines of communication became more exposed. Japan was fighting precisely the war which least suited its material resources, a prolonged and costly battle of attrition beyond easy reach of its supply system.

This drove Japan, as a matter of routine, to take great and often costly risks, such as shipping divisions to the most forward ports under its control, and into the maw of enemy air and naval strength, or operating hundreds of aircraft from the same base, all vulnerable to one massive onslaught. Japan's fighting and logistical services entered a vicious circle, which progressively weakened their ability to fight anywhere at all. For as both sides strained their muscles to strangle one another in New Guinea and the Solomon Islands, Allied forces began to *slash the exposed limbs of Japanese logistics*. The southwest Pacific witnessed a terrible campaign of maritime interdiction, rendered even more deadly and one-sided by Japan's failure to retaliate in kind and its incompetence in convoy escort. These engagements eroded still further the relative ability of Japanese logistics to support a prolonged war of attrition. By early 1944, exactly as the IJA adopted a strategy of human (though not material) attrition. Japan had lost the naval and air strength needed to back it. Japan was on the verge of losing any use from every single soldier and ton of supplies on its forward defensive perimeter from the Marshall Islands to Burma.

These American engagements also directly shaped operations: the battle of the Bismarck Sea drowned not merely a division but any hope for Japanese strategic intiative in New Guinea. Similarly, pre-emptive airstrikes wrecked Japan's forward airbases on that island, largely because the Japanese had failed to develop an effective early warning system for air defense or to use properly its few radar sets. This crippled Japan's air performance and increased the scale of its aircraft losses.

Between August 1942 and February 1944 the great American victories in the Pacific lay in the war of attrition. No doubt some such success would have occurred in a world without Ultra but, as Drea demonstrates, in the world as it was these triumphs did rest on the foundation of signals intelligence. As a result, small American forces were used with optimum effect against their logistical and air targets. A few hundred aircraft and a few dozen submarines on station did not have to hunt tiny and elusive targets in an area of a million square miles. The location, strength and intentions of those targets were known; little American effort was wasted and few opportunities were missed. As the generator of both the economy and the concentration of force, intelligence came close to reducing uncertainty. It allowed precisely intended actions to be executed precisely as intended. By ensuring that all available American naval and air power would be used effectively against the enemy, it turned aggressive risk-taking into calculated risk. MacArthur's air force commander, General George Kenney, was no better than SWPAC in dealing with ambiguous information. Yet Kenney received vast amounts of 'hot' and 'hard' material from Ultra, which he trusted and used in one of the most successful campaigns of interdiction and the acquisition of air supremacy ever waged by any air force. Few commanders have married intelligence so well to operations and with such effect.

Events on the ground in New Guinea during 1942–43 took a different form. Despite the breathtaking self-sacrifice of Japanese soldiers and their formidable military qualities, the IJA lost more men than the Allies in every major engagement. Nor could it match Allied material resources. On the tactical and operational levels, however, the IJA's attacks often caught the Allies by surprise, making them fight in areas where their advantages in material could not easily be deployed. When on the defensive, the IJA forced the Allies to launch slow and costly battles of attrition. All this offered Japan the chance for perpetual check in New Guinea. If forced to fight on the IJA's terms, the limited resources of SWPAC could do no more than slowly reconquer a wilderness. This is exactly what happened during the middle game of this campaign. Whether the Allies used their armies clumsily, as under MacArthur in the summer of 1942, or with skill, as under Blamey in the spring and summer of 1943, they were not pursuing a winning strategy. Instead, they played directly into the IJA's policy of delay, human attrition and the grinding of American will. Nor was Ultra of great value in these operations. In technical terms, it was in a primitive state, producing little material directly relevant to land operations – material which, in any event, was difficult to interpret. It sometimes illuminated the enemy's strategic redeployments to the theater but MacArthur's armies (unlike his air forces) were in no position to act on that material. Given the nature of the terrain and the naval and air balance between the adversaries, Allied land forces had a limited range of options, all of which led to a pattern of stereotyped operations along narrow and predictable channels of advance. So long as this continued, SWPAC could never take center-stage in American strategy nor could Ultra offer more than marginal assistance on the field of battle.

At the turn of 1944 all of these things changed, for a brief but crucial period, as did MacArthur as military commander. His record in the Pacific before April 1944 and after November 1944 was mixed. On the one hand, he evinced aggression, willpower, a gambler's instinct and a flair for amphibious operations. Against that stood his penchants for misreading enemy intentions, for issuing unrealistic orders and for attacking directly where the enemy was strongest; his operations on land, moreover, tended to become bogged down in battles of attrition, a style of operations in which he was not gifted. Conversely, in mid-1944 MacArthur conducted an impressive campaign which turned his command from the periphery to the center of Allied strategy in the Pacific. Not coincidentally, this occurred precisely when Ultra finally and suddenly penetrated into the IJA's operational traffic. That was a necessary cause for MacArthur's successes, but it was not a sufficient one. The side with superior intelligence also possessed greater military power and favorable opportunities to fight a war of maneuver. The war of attrition had gutted Japanese naval and air resources, the bulwarks needed to brace its defense perimeters against major blows. American military power heavily outweighed that of Japan in quantity and quality, while the breaching of the Bismarck barrier and the cracking of the shield of the Marshall Islands opened a 1,000-mile line of sea room for a campaign of maneuver toward Japan's inner defense perimeter. On New Guinea the IJA remained formidable, with roughly as many soldiers as SWPAC possessed, but it was strong in only a few locations along a 1,000-mile coastline. Elsewhere it was not merely weak but almost literally non-existent. Its logistical structure consisted of one major supply centre and a handful of small ones, linked by coastal shipping and portage on human backs through roadless bush. This system could sustain the forward crust of the IJA against the heaviest direct thrust which SWPAC could launch, but just one deep and heavy amphibious blow might shatter it beyond hope of repair. In 1944, unlike 1943, such blows could be launched. Japanese air forces on New Guinea might endanger or annihilate amphibious operations but they were far weaker than Kenney's command and, as ever, were vulnerable to pre-emptive airstrikes. Admiral Halsey's Third Fleet in the south/central Pacific generally tied down the IJN. The latter, however, could still despatch naval forces greater than those under MacArthur's command for short periods of time to the waters around the northern half of the New Guinea theater.

The erosion of Japanese naval and air resources, combined with the rise in his own logistical and amphibious capabilities, gave MacArthur the opportunity to strike long and heavy blows at Japanese positions. The potential range of his amphibious assaults increased from 80 to 800 miles from his main base. In principle, a repetitive pattern of stereotyped operations in predictable sectors of attack could be avoided. But in practice, as Drea demonstrates, without Ultra MacArthur would not have used these material resources with maximum effect; he might not even have understood the new range of options open to him. Certainly in early 1944, before the intervention of Ultra, MacArthur still pursued a stereotyped line of operations: to strike directly at Hansa Bay, the enemy's main line of resistance. This would have played straight into Japan's hands. The IJA treated east-central New Guinea as a forward glacis on which to delay the enemy and to weaken its strength. Meanwhile it prepared to make north-western New Guinea a bastion for part of the inner defensive sector, the Philippines. As Drea emphasizes, a direct attack on the outer crust of Japanese defences at Hansa Bay might well have failed and, had it done so, the southwest Pacific theater would have been completely marginalized.

Once Ultra of a self-evident nature uncovered Japanese deployments on New Guinea, in particular, the strength of its forward crust at Hansa Bay and the weakness of its main supply base at Hollandia, MacArthur jettisoned his plans. To precise knowledge and superior force he added aggressive command and the calculated pursuit of great gains at the risk of moderate losses. In March–April 1944 Kenney's forces demolished enemy air strength in central New Guinea, followed immediately by the seaborne seizure of Hollandia. This shattered Japanese defenses in the southwest Pacific and split its forces on New Guinea into two halves. Over the next two months a fast and aggressive amphibious campaign seized north-western New Guinea before Japanese defenses could be solidified, while several IJN counter-attacks were parried. During the four months between April to July 1944, MacArthur leapt five times further than in the previous 24 months: his forces annihilated or isolated five IJA divisions while suffering small American losses and smashed in Japan's external defensive perimeter – just as Halsey's forces were doing 1,000 miles to the north. MacArthur also seized bases for the assault on the Philippines, the projected date of which was advanced by five months. Then the attack on the relatively weak garrison of Leyte in the Philippines during October 1944 caught Japan by surprise. Ultimately, the American hammer smashed three divisions and what remained of Japan's regular naval and air forces on the anvil of Leyte. This gave MacArthur the opportunity to render irrelevant and bypass much of the inner perimeter of Japanese defenses, the Philippines. In any case, he could now directly attack the most vital and vulnerable components of the Japanese Empire – its line of maritime communications from Singapore to Tokyo, and the Japanese home islands.

These operations and MacArthur's entire strategy of bypassing enemy strongpoints and concentrating his power against weakly held but strategically vital areas that could serve as forward logistical and air bases, rested on intelligence, in particular, on Ultra. In the battle for Leyte, Ultra also allowed American airforces to smash several units and portions of the Japanese 23rd and 26th Divisions at sea, thus crippling the defense of that island and the quality of several good formations. Yet Ultra increasingly lost that function and this value once American forces entered the Philippines. Tough fighting could no longer be avoided, since strong garrisons held each position in the innermost perimeter of Japanese defenses. MacArthur refused to bypass most of the Philippines but insisted on assaulting many of its main islands, because he viewed the conquest of the Philippines as an aim rather than a means: and an aim defined far less by the dictates of strategy than his own personal prestige. The logic of the battlefield and of MacArthur threw American forces directly against centers of Japanese strength, instead of around them. This situation was like that in New Guinea during 1943, with one difference – given a vast increase in their firepower, American forces ground down the IJA more quickly and cheaply than before. By 1945 the Japanese command on Okinawa calculated that every American division possessed five times more firepower than any Japanese one.[52] Even so, MacArthur's operations cost time and American lives, and many of his assaults were unnecessary for the advance upon Japan.

Simultaneously, these battles required the support of precisely that sort of intelligence which SWPAC could least provide. MacArthur's great campaign of March–October 1944 had rested on the basis of self-evident Ultra. It had required exactly the material which Ultra was best able to acquire and SWPAC to understand and use, on the enemy's strategic intentions, perceptions, dispositions and redeployments. Intelligence on these strategic issues could differentiate enemy strongpoints from centers of weakness. If SWPAC could determine where the enemy was weak, it did not need precise intelligence on the composition of garrisons which were known to be strong, so long as it proposed to bypass them. This was fortunate, since even in mid-1944 its assessments of such operational intelligence were mixed in quality. The lacunae in Ultra and errors in its analysis led to gross miscalculations of the strength of several garrisons which American forces did have to attack, failures in assessing the maritime movement of some Japanese reinforcements, especially its First Division, to Leyte, and a misunderstanding of the enemy's intentions before the divisional strength attack at Aitepe in New Guinea.

Once the strategy of bypassing was abandoned, and American forces were condemned to attack Japanese strength head-on, these problems in the acquisition and assessment of intelligence acquired more and more importance. With the near-annihilation of the IJN and of Japan's ability to move soldiers and supplies by sea, cryptanalytical access to traffic about maritime matters became irrelevant. Yet previously that had been the main source of signals intelligence and of self-evident Ultra, the material which SWPAC used most effectively. While the IJA's operational codes yielded far more material than before, this could not be used in a straightforward fashion. In essence, this traffic provided opaque Ultra, SWPAC's traditional weak point, about the least comprehensible aspect of the IJA, its order of battle and the strength of its formations. Moreover, there must almost always be gaps in the material provided by any single source of intelligence. Continually from November 1944 SWPAC misunderstood these issues, while MacArthur ignored Ultra whenever it did not say what he wished to hear. Not only this, to a large extent his strategy threw away the potential advantages offered by Ultra.

When American forces entered the Philippines Ultra became less and less useful. Misled by a combination of fragmentary intelligence, especially regarding the strength of smaller formations and support troops, and an analysis which rested on wishful thinking, Willoughby underestimated the number of Japanese defenders on Luzon in the Philippines by 50 per cent, or 130,000 soldiers. SWPAC's planning for the invasion of Luzon rested on this assumption. MacArthur in particular wished to fight a rapid and risky campaign which would have played straight into the hands of a strong, effectively deployed and well-led enemy. Fortunately, working from exactly the same data, Krueger's headquarters assessed Japanese strength accurately and far better than SWPAC did. Krueger, not MacArthur, ran the battle and he did so through a broad and cautious advance which took Japanese power properly into account. This reduced to a minimum American casualties but also the speed of the campaign. On Luzon as on Okinawa, isolated and outnumbered IJA forces substantially reduced the American strength available to attack the Japanese home islands. By 1944, alone among the Axis armed services, the fighting quality of the IJA had not generally collapsed from its peak of 1942. A strategy of bypassing rendered most of its strengths irrelevant and exploited all of its weaknesses. Once bypassing was abandoned, the abiding power of the IJA could again support Japan's strategy of delay and attrition of American manpower and willpower.

Gaps in Ultra and mistakes in its interpretation did not hamper the invasion of Luzon but neither did signals intelligence particularly help it. Ultra showed Krueger that he must fight a careful and cautious campaign, but this was his predilection in any case, and not an unreasonable one given the known quality and characteristics of the IJA. Ultra certainly had little effect on his operations once ashore.

In Operation Olympic, the projected invasion of Kyushu, Ultra was more successful. It was the only source of strategic intelligence available to American planners and in some ways a strikingly effective source. It located 13 of the 14 divisions in Kyushu and uncovered the IJA's entire defensive strategy. Conversely, Ultra did not define the strength of these units. Thus Willoughby underestimated the number of

Japanese soldiers on Kyushu by some 40 per cent. MacArthur rejected this calculation as defeatist and held that Japanese forces were even smaller than Willoughby supposed. Fortunately, the plan of campaign which rested on this optimism was not put to the test, largely because of the indirect influence of Ultra.

Ultra also shaped the politics of American strategy at the theater level throughout 1944–45. It was not the only factor at play, but until Ultra is accounted for, the decisions to invade the Philippines and to use the atomic bomb will be misunderstood.[53] During 1942–43 SWPAC was a command of tertiary importance. The operations of 1944 made the territory under its control a plausible base for the assault on the Japanese home islands: and MacArthur immediately brought the Philippines to the forefront of American strategy. While Chester Nimitz and 'Bull' Halsey, potential rivals for dominance over American strategy in the Pacific, accepted MacArthur's case, the Joint Chiefs of Staff were less accommodating. Here Ultra became MacArthur's trump card. During the debate over the future of American strategy between June and August 1944, assessments of Japanese capabilities and intentions rested primarily on Ultra. MacArthur's operations of the period, his drafting and redrafting of plans for the invasion of the Philippines, were shaped by Ultra not only in operational terms but in political ones – by the need to overcome the arguments, often themselves derived from interpretations of Ultra, put forward by opponents to his strategy. Conversely, Ultra may have saved Americans and Japanese alike from paying the price on Kyushu for MacArthur's ambition to command the largest amphibious assault the world had ever seen.

Ultra was fundamental to the American decision to use the atomic bomb: indeed, Edward Drea has demolished the view that intelligence should have led Washington away from any use of that weapon. The best source of strategic intelligence available to American decision-makers in the summer of 1945 was not 'Magic' but Ultra. Magic demonstrated – indirectly – the intentions of the liberal Japanese faction, which had routinely lost power struggles in Tokyo for 15 years. It revealed that even they pursued peace terms which the United States would not accept. Ultra revealed – directly – the capabilities of the IJA and the intentions of the faction which dominated Japanese policy. It showed that the IJA intended to make Kyushu a second Okinawa and fight to the death, taking tens of thousands of Americans killed and hundreds of thousands of wounded with them, not to mention millions of Japanese. An assessment of the intelligence record as a whole – as against mere Magic – makes the attitudes of American decision-makers towards the atomic bomb more explicable. Ultra directly pressed the

United States towards the use of the atomic bomb and away from a conventional amphibious assault.

Viewed with hindsight, the Pacific war was a classic case of military overkill. Japan was so outweighed by Western technology and American industry that a single-handed Japanese victory was impossible. Even its survival as an independent state was unlikely. Only if Germany had won the war in Europe and come to rescue its ally (which, in turn, might have been possible only if Japan had joined in an attack on Russia during 1941 instead of starting the Pacific War in the first place) was any kind of Japanese victory conceivable. So, even though signals intelligence clearly was involved in Japan's defeat, obvious questions remain. How important was Ultra? Was it necessary to victory? Could the war have been won without it and in what way? While these questions cannot receive a definitive answer, they can be addressed through three arguments.

First, in 1942 Ultra was a necessary although not a sufficient cause for the elimination of the IJN's superiority at sea in the battle of Midway – luck, Japanese errors, American strength and Chester Nimitz were of equal significance. Then, by shaping the American decision to invade Guadalcanal, traffic analysis helped to lead Japan toward a campaign of material attrition in the south and southwest Pacific.[54] Ultimately these events significantly hastened the withering of Japanese military power.

Second, in 1944 Ultra was necessary to the strategy of bypassing, which short-circuited Japan's hopes to grind down American will through human attrition. This Japanese strategy was not necessarily foredoomed. Even accounting for the desire to avenge Pearl Harbor, the United States would not have had to pay an unlimited price simply to annihilate Japan. Had conventional forces been the only kind involved, the United States might have been ready to accept a more favorable kind of negotiated peace than Japan in effect received. After all, in Korea and Vietnam between 1950 and 1975, states with no more firepower than Japan and a similar style of military operations so eroded American will and killed American soldiers as to make the United States tolerate a military and diplomatic stalemate. The United States suffered heavy losses on New Guinea and Tarawa in 1943, Leyte in 1944, Luzon, Okinawa and Iwo Jima during 1945. This butcher's bill would have been far higher had the United States been unable to leap through Japan's defense perimeters by attacking only a few and weakly held islands. Of course, even then it can never be established whether the United States would have refused to pay the bill for the simple pleasure of making Japan surrender unconditionally.

Third, however, that argument is not merely hypothetical but irrelevant. Even had the American Army been ground to a halt on the field of battle, atomic weapons probably would have still forced Japan to surrender – and Pearl Harbor ensured that these weapons would be used. Thus, signals intelligence clearly was not necessary to American victory in the Pacific war. Only if Germany had won the struggle in Europe and the United States been forced into a simultaneous two-ocean war might Ultra have become the straw which broke the back of Japan. On the other hand, very few single factors are absolutely essential for victory in a prolonged war of attrition and signals intelligence clearly was a major contributing cause to American victory. It allowed the United States to defeat Japan far more speedily and at a lower cost. This, of course, was exactly what Ultra allowed the Western Allies to do against Germany in Europe at the same time.[55]

This analysis rests largely on the narrative of Edward Drea. He has also provided several general observations about Ultra and MacArthur which contribute to an examination of intelligence and military operations. Contrary to Drea's own expectations, Ultra never shaped MacArthur's strategy; it rarely affected the formulation (as against the execution) even of his major operations. Whatever intelligence indicated, MacArthur's aims never changed: to make his command the center of the American effort in the Pacific, to retake the Philippines and to win the war through an invasion of Japan. These objectives, Drea insists, stemmed from vanity no less than rationality, although to some degree they were defensible on strategic grounds. For MacArthur, Ultra was a tool to achieve his own aims rather than to help him choose among them. He used intelligence when this helped him further his objectives and ignored it when that would not. While this is conventional for many commanders, it is not always so. MacArthur falls into the group of generals in the twentieth century who were least likely to change their aims or means because of intelligence. Finally, Drea offers a brief and tantalizing statement about the effect of intelligence on commanders of specific personality types: 'ULTRA appears to have reinforced basic personality traits. It convinced forceful commanders to take risks and push forward, just as it persuaded prudent ones to go even more slowly.'[56] From a study of this generalization, combined with the use of the intelligence record as a mirror of generalship, one can link together different issues: several generals' use of Ultra and their style of command, the relationship between Ultra and uncertainty, and the role of intelligence in war. All of these issues, in turn, illuminate matters fundamental to Clausewitz's arguments and to the study of war: the importance of intuition and determination in thought, of boldness in operations, the nature of understanding, command and action in war.

INTELLIGENCE AND MODERN WAR

When his actions are compared with his knowledge, Kenney stands out as a bold, flexible, quick-thinking and far-sighted commander. Relatively superior intelligence of a high absolute order was fundamental to his success, yet he acted on knowledge with skill, resolution and audacity. Here is a commander who embodies what generally would be regarded as a superb union of knowledge, boldness, superior intellect and cunning. Krueger and, to a lesser extent, Blamey fell into the psychological category which Clausewitz regarded as the norm: the cautious commander who wanted reinforcements against uncertainty. And of course overwhelming superiority is the most effective guarantee against uncertainty – not surprisingly, the first rule in Clausewitz's art of war is to be as strong as possible, especially at the decisive point.[57] Yet one must not separate psychological characteristics from their operational and doctrinal context. Caution, if guided by knowledge or intuition, as opposed to being merely an irrational response to uncertainty, could be a valuable characteristic against an enemy whose offensive operations were stereotypically aggressive and whose defensive positions were well prepared. To adopt an aggressive but unbalanced defense against a Japanese attack was to give the IJA its greatest chance to play its strong cards of speed and maneuver and to avoid the Allied trump card of prepared firepower; conversely, caution allied to knowledge, intuition or cunning could force the IJA to attack Allied strength head-on. Similarly, for an Allied commander to launch aggressive and unprepared assaults against thick Japanese defensive positions would be to throw away his trump card and to gamble on the weak card of the ability to tolerate losses. Daring in operational maneuver through such matters as outflanking or bypassing was often, though not always, an effective way to negate the IJA strength – it certainly led to the greatest of Allied successes against Japan. When forced to attack prepared Japanese positions, however, boldness in tactics and operations was almost by definition recklessness, for precisely the same reasons that it had been on the western front during 1915–17. Set-piece battles were the rational response to this situation. Caution rather then boldness might best allow a commander to exploit the known offensive and defensive characteristics of the IJA.

For Blamey and Krueger caution, combined with skill in set-piece operations, did pay great strategic dividends. Blamey's plan for the drive on Lae and Salamaua in New Guinea between June and September 1943, for example, deliberately eschewed the boldness preferred by MacArthur, and properly so. As Blamey calculated, by instinct rather than knowledge, to advance further than Nassau Bay would be to attack the IJA where it was strongest. By stopping at Nassau Bay, he could force the IJA to advance and attack him, out of effective range of its own air support and into the maws of his defensive firepower. All this, in turn, would weaken the IJA's rear defenses, ease the success of another and deeper amphibious strike and, when that happened, trap more enemy forces.

This plan worked as planned, which is unusual in war. It was similar to a successful approach adopted by the Canadian Corps against the German Army in the battle of Hill 70 in 1917 and by the Canadian and Australian Corps at Amiens in 1918 – proof that caution, the use of the operational offensive combined with the tactical defensive, and set-piece battles guided by knowledge and intuition, have a proven track record against armies which institutionalize aggressive risk-taking alongside defensive systems relying on tactical or operational counterattack.[58] This approach to operations made Dominion forces, division for division, the Germans' most effective enemies in both world wars. It could easily be used against any enemy which followed German defensive systems, including the IJA. At Lae/Salamaua, Blamey consciously applied the style of operations which the Dominion armies had used to smash the Germans in 1917–18 and which they were using to the same purpose in the Middle East and Europe. His understanding stemmed both from an instinctive grasp of how the IJA fought and of how set-piece battles were conducted, and from better intelligence than Clausewitz would have expected, though this stood far from the standard of perfection. Ultra informed the Allies of the overall strength of Japanese forces on New Guinea, but their specific deployment remained very unclear. Hence Blamey's strategy rested one third on information, one third on cunning and one third on a Clausewitzian technique to manage uncertainty. The attack on Nassau Bay acted as a magnet for the IJA, and thus uncovered its deployment in the forward area; this also threw the enemy off balance. Caution 'governed by superior intellect' overcame uncertainty and an aggressive enemy.

Krueger lacked Blamey's brilliance. He was a cautious, methodical and deliberate commander, perhaps more so than is justifiable, but his approach, none the less, was effective. He cannot be faulted for adopting a bad means to deal with the IJA, merely for not choosing the best, and even this criticism is of dubious accuracy. During the course of his operations Ultra had reached a state of technical maturity while Krueger also established an able intelligence staff. Professional assessments of intelligence shaped his operational aims and means no less than was true of Kenney. The intelligence record explains and to a large extent justifies the controversial decisions to adopt slow and methodical means to clear both Leyte and Luzon. Krueger's caution stemmed from two distinct but related factors, a personality reflex and a rational response to specific circumstances. Headlong charges against deep and prepared Japanese defenses would produce far more American casualties for no better results in battle than prepared attacks, and perhaps for no quicker results.

The mirror of Ultra offers an odd reflection of MacArthur, one which should be considered by students of his operations during the Korean War. He had a closed mind regarding aims. He was remarkably reluctant to change even his means once he had defined them. When on the defensive, in conditions of uncertainty, against an enemy with the initiative and superior force, as between June and August 1942 during the period surrounding the Japanese invasion of Buna and Milne Bay, MacArthur could be a cautious, even a passive, commander. When gripped completely in the toils of uncertainty and facing an enemy with overwhelming power and virtually complete initiative, as on the Philippines in the hours immediately following the outbreak of the Pacific war, MacArthur could be utterly paralysed. He was at his worst in adversity, and at his best when he possessed superior force and the initiative. Then, daring and tenacity of purpose were MacArthur's cardinal characteristics – if not always desirable ones, as Blamey appreciated at Salamaua and Krueger on Luzon. As MacArthur wrote in February 1942:

> Counsels of timidity based upon theories of safety first will not win against such an aggressive and audacious adversary as Japan . . . The only way to beat him is to fight him incessantly. Combat must not be avoided but must be sought so that the ultimate policy of attrition can at once become effective. No matter what the theoretical odds may be against us, if we fight him we will beat him.[59]

MacArthur fits the crude version of Clausewitz's 'military genius', with the emphasis upon the determination, willpower and aggressive risk-taking of the commander. In fact, however, with MacArthur as with Douglas Haig, the vulgarized version turned the model on its head. Strength of character, boldness and determination became closedmindedness, obstinacy and lack of originality. The value of MacArthur's psychological characteristics as commander turned on intelligence and military conditions. Given the knowledge and the chance to maneuver away from the enemy's strength and toward its weakness, MacArthur's aggressiveness produced impressive actions. Without that knowledge (or the willingness to use it), without that opportunity and will, MacArthur's weaknesses as a commander came to the fore and aggression tended to become liability. Time after time he took great risks to win the biggest stakes, a kind of mentality which must inevitably lead a commander to a crushing defeat. Time after time he preferred to assault the enemy where it was strongest. Ultra staved off the potential consequences of these weaknesses, but they had their inevitable result along the Yalu River in 1950.

A combination of superior power, room to maneuver, good intelligence and luck was necessary to turn MacArthur's daring to use. Of these four, the first two were necessary in individual terms. The latter two were collectively necessary but on its own each was merely a contributing factor. MacArthur could live without luck so long as he had intelligence. He could often live without knowledge so long as all of these other conditions held true, but intelligence always made his operations easier and it was necessary for his great successes between March and November 1944. MacArthur also acted on secret information with resolution and speed – when he wanted to believe it. Unlike Kenney and Krueger, however, he was not an intelligence-driven commander. In effect, he acted on Clausewitz's '*imperative principle*' – he changed his first opinion only when forced to do so by clear conviction, and only when irrefutable evidence pointed toward a better means to achieve his prime directive, and he never changed his aims. With MacArthur, the better and more trusted the knowledge, the more daring and determined the action: the less certain the situation, the more pedestrian the response. Yet this was just a tendency, a strong but not an invariable pattern. MacArthur finally turned the Bismarck barricade in late 1943 through the invasion of Arawe, which rested on a gamble rather than knowledge. This was an instance of aggressive risk-taking that Clausewitz himself would most likely have regarded as bold.

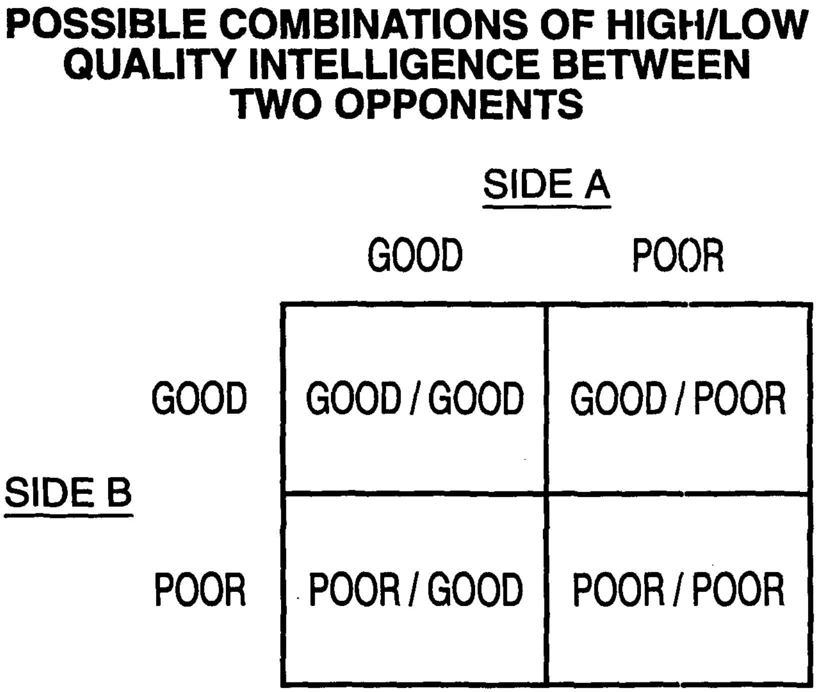
The intelligence record determines some of the characteristics of MacArthur and his lieutenants and also indicates their quality. It enhances Kenney's reputation which needed little refurbishing, while substantially rehabilitating that of Krueger. It also shows Blamey to have been a shrewd commander, a surprisingly able one who understood both the enemy and the art of war in his theater. MacArthur does not come out of the study at the level indicated by Drea – as one of the four or five best commanders in the war – but he certainly stands out as a good and gifted general, of mixed qualities and achievements. The intelligence record demolishes the hero-worship and self-mythologization of earlier works along with the uncomplimentary assessments of the recent revisionist literature. It shows him warts and all, as a far from average commander – weak in some ways, strong in others. A reading of Drea also leads to a surprising conclusion: while intelligence staffs notably improved their ability to handle Ultra, there was no learning curve in its use by MacArthur and his lieutenants. They started as well as they finished, as regards Ultra of both categories, fragmentary and finished material. Indeed, MacArthur handled intelligence in Korea exactly as he had in the Pacific. In particular, during the drive to the Yalu his refusal to consider intelligence which challenged the aims he had laid down, and his use of intelligence in policy debates with his superiors, were identical to his practice of 1945. The record also sustains part of Drea's argument that Ultra reinforced 'the basic personality traits' of specific commanders. It certainly led aggressive risk-takers to become even more audacious but it led cautious ones to become not more prudent but rather more calculated. This observation has significance for the theory of command and decision-making in war.

INTELLIGENCE, OPERATIONS AND THE FUTURE OF WAR

Some provisional observations may be drawn about the modern relationship between intelligence and uncertainty, and its effect on the art of war, the psychology of command and the nature of and need for military genius. These conclusions pertain specifically to a situation where superior and knowing forces fight inferior and ignorant enemies, who adopt a vulgarized version of the Clausewitzian formula for managing uncertainty. It also applies to cases where one side possesses intelligence services which are (a) of an absolutely good quality and (b) superior to those of the enemy. These conclusions do not necessarily apply to different circumstances. When both sides possess equal and effective intelligence, for example, the result may be the military equivalent of chess. Alternatively, when intelligence of such quality and equality is derived from sources that can easily be destroyed if only they are known to exist – as would have been true of many types of Ultra – the same situation may lead each side simultaneously to blind the other and plunge the battlefield back into darkness. Similarly, when both sides possess equally poor intelligence, Clausewitz's predictions may once again pertain. However, a mediocre intelligence service when con-

FIGURE 1

POSSIBLE COMBINATIONS OF HIGH/LOW QUALITY INTELLIGENCE BETWEEN TWO OPPONENTS



fronting a poor one may give its master a remarkable edge. One need not move far from uncertainty before Clausewitz's rules cease to apply with certainty.

In the case at hand, when *intelligence helps the stronger side but not* the weaker, the latter fights in the fog of war, the former merely in mist. The debilitating effect of confusion and uncertainty is reduced for one but only one army. Only reduced – never eliminated. Intelligence can never achieve complete and absolute perfection; no sane commander will ever expect it to be so. As E.T. Williams, the head of intelligence for the British 21st Army Group, wrote about his experience with Ultra:

> Perfect Intelligence in war must of necessity be out-of-date and therefore cease to be perfect. We deal with partial and outmoded sources from which we attempt to compose an intelligible appreciation having regard to the rules of evidence and our soldierly training and which we must be prepared constantly to revise as

> new evidence emerges. We deal not with the true but with the likely. Speed is therefore the essence of the matter.[60]

Any single piece of information can remain accurate for only a certain period of time: completeness in intelligence requires the collation of many such pieces: and during the time required to achieve perfection in that process, some of these pieces cease to be accurate. Given the dialectical and paradoxical nature of strategy and intelligence, moreover, good information well understood about a phenomenon is likely to make commanders take actions which change the phenomenon and thus render imperfect the information, its assessment and the actions flowing from them.

By definition, even with excellent intelligence, friction and the reciprocal nature of war ensure that uncertainty will still exist, if in a different form and, perhaps, to a lesser degree. Nor are intelligence and uncertainty linked in a simple linear relationship. As one first moves a certain distance away from the pole of ignorance, more intelligence may create more certainty; but as one moves closer still to the pole of knowledge, the greater the volume of information, again the greater the uncertainty: indeed, a new kind of uncertainty will be produced. Nor can intelligence ever be a universal panacea in war. Its value will always vary with the operational circumstances. Good intelligence well understood may be useless: a good army with bad intelligence may well defeat a bad army with good intelligence.

None the less, such instances as MacArthur's operations of 1944, Kenney's air campaigns and the Allied amphibious assaults on Europe, all point to one thing. Against a bold, powerful but ignorant enemy, especially one whose forces are unbalanced or taking substantial risks, a daring and powerful force reinforced by knowledge can strike far more precise blows than Clausewitz would have dreamed possible. The greater degree of precision will render these blows all the more deadly. This may also be true of inferior but informed forces, as shown by Nimitz's decision to gamble on knowledge at Midway. Yet, as Blamey, Krueger and Montgomery at El Alamein and Medennine demonstrate, the frequency and the success of calculated styles of command have also risen, particularly when attacking a strong and effectively deployed enemy.

Clausewitz's rules of thumb and estimates of probable outcomes do not completely fit modern circumstances because some of his assumptions are no longer valid. Far more frequently than he expected, for example, generals were able to strike precisely intended blows precisely as intended and to surprise their enemies in specific ways through deliberate intent. Moreover, a larger fraction of generals seems to have been able to function effectively in conditions of knowledge than of ignorance. Many of these generals may not have been military geniuses, but perhaps military genius is less fundamental to command in circumstances of knowledge: perhaps then military competence augmented by good intelligence will suffice. While not all men can be good commanders, some previously barred from that status now could achieve success on the battlefield when the heightened power of intelligence reduced the scale of uncertainty and the density of the fog of war. This reduced some – not all – of the friction of war; arguably, so too did other improvements in the workings of military technology and organization, such as communication systems. In particular, during the era of Clausewitz, the commander almost literally did perform the entire function of command: his staff was nothing but a small secretariat. Since then, command has become the *joint function of a man and an institution*: the staff has become a bureaucracy and the commander anything from chairman of the board to CEO. Room remains for the freedom of action of the commander, the influence of his personality and the exercise of his genius, especially at a tactical and operational level, but these factors are restrained. The corresponding loss in potential drive and creativity is balanced by the improved general efficiency of the military machine. This has changed the nature of the personality which understands and acts on intelligence: once it was a single mind, now it is one mind harnessed in different ways to an institution. In some cases, the commander becomes an executive implementing a collective decision and his personality becomes less relevant to command; in others, his personality remains the dominant factor at play.

The heightened power of intelligence altered the intellectual faculties which Clausewitz thought necessary for command. He regarded 'presence of mind' as valuable, for example, because it alone allowed immediate and competent decisions when some of the fog of war suddenly lifted. But for that group of commanders who knew in advance what lay behind that patch of fog, this faculty became less significant than the ability to act effectively on such knowledge. Informed intuition probably remained essential to military genius, but, perhaps, not to military competence so long as that faculty was augmented by intelligence. Above all, the heightened value of intelligence sapped the value of determination (in Clausewitz's sense) as a factor in military genius. That commander who dogmatically followed Clausewitz's '*imperative principle*' of sticking tenaciously to first opinion despite contradictory material was a fool. Many more commanders had 'adequate grounds' for action than Clausewitz thought possible, and therefore had less need for determination (in his sense). When time was short, of course, presence of mind remained as crucial: when battles became fluid, and very often good intelligence and real-time communications systems failed, then determination (in Clausewitz's sense) remained vital. None the less, qualities fundamental to command in the age of uncertainty often proved counter-productive in the era of Ultra – of good intelligence and real-time communication. Indeed, military genius itself may actually be counter-productive in an era of excellent and reliable intelligence. Geniuses do not need great knowledge to perform effectively. They often make a fetish of faith in their intuition.[61] Military competents need all the help they can get and often know that fact. They have far more than military geniuses to gain from reducing the environment which makes geniuses so formidable – uncertainty – and this obviously can be done through a use of intelligence no less than through the concentration of superior power. Greater intelligence, that is, cuts at the roots of the superiority of the military genius – it may altogether eliminate that edge.

Nor is that the end of the story. Intelligence affects another important condition of war. The effect of 'fortune' – of unforeseen, unforeseeable and unintended matters – was central to all classical theories of strategy. Thucydides and Machiavelli, Clausewitz and Napoleon, all regarded luck as fundamental to success or failure in war. Many premodern commentators implied that strokes of fortune were caused by the intentional intervention of providence or else somehow were the natural and consistent product of a commander's character. Thus, so goes the story, when considering whether to make a man a general, Napoleon asked one question: 'Is he lucky?'. Clausewitz, conversely, stripped connotations of intention and the consequences of character from his idea of 'chance'. To him, chance meant nothing more or less than random events produced by the interplay between a huge range of unforeseeable and uncontrollable factors.

Precisely these issues have changed in the modern age of better intelligence, real-time communication, and powerful staffs. What once were random actions now often can be predicted and controlled. Armies can foresee far more than before, and execute their intentions far better. Thus, the effect of 'fortune' on war has declined along with its plausibility as an excuse for defeat. Not that luck can ever vanish from war. It will particularly affect the tactical and lower operational levels, often as much as it has always done. On the plane of higher operations and strategy, however, its significance will decline. Luck will remain a factor in calculated risks, though far less important than in the era of uncertainty. When two armies with informed but opposing intentions clash, moreover, much that they know will change and so will their ability to execute their intentions. Above all, where one side has better intelligence, it will rely on knowledge while the fate of its foe will hinge on chance. Intelligence can directly replace luck. 'Fortune', so long queen of the battlefield, has been dethroned by a pawn.

The heightened power of intelligence in the Second World War did not change the relationship between generals exemplifying Clausewitz's two kinds of unregulated temperament: aggressive risk-takers remained superior to timid generals. Intelligence, however, did increase the superiority over these two of every case of temperament guided by reason – it provided a more solid and rational basis for audacity governed by superior intellect or cunning, and deliberate caution. It also increased the frequency of these cases. Krueger and perhaps even Montgomery or Blamey might have been timid generals in a different age of uncertainty and intelligence, but in the Second World War they were calculating ones. Hence, aggressive risk-taking became a conditional virtue. Sometimes it produced all the rewards promised by boldness (in Clausewitz's sense). Aggressive risk-taking, however, was automatically and without warning converted to recklessness whenever it confronted boldness or deliberate caution guided by intelligence. Clausewitz himself noted: 'Even in daring there can be method and caution; but here they are measured by a different standard.' They must be measured by a different standard again in conditions of greater certainty. Thus, the exceptions to Clausewitz's generalizations about audacity became increasingly common – so much so as to challenge the rule.

All told, between 1914 and 1945 better intelligence reduced the value of certain characteristics in the Clausewitzian equation, particularly determination and nerve, and increased that of others, especially calculation and the ability to act on sure knowledge in a precise way.[62] Certain gifts of character declined in value, some others of intellect and technique rose. Clausewitz most prized the gifts which allowed commanders to act without knowledge, but all these gifts have different values when one can act on knowledge. His prescriptions were formulated to suit the intelligent commander without intelligence: they had to be revised to suit the intelligent (or the competent) commander with intelligence.

But this situation may have changed again. The world wars of this century were, perhaps, that period when the ability of armies to acquire, assess and act on intelligence was at its peak in history. The relationship between uncertainty and intelligence may have changed as much since

FIGURE 2

THE EVOLUTION OF THE ROLE OF INTELLIGENCE IN MILITARY OPERATIONS AND WAR

	1800*	1914-1945	PRESENT AND FUTURE
THE COMMANDER'S INTEREST IN INTELLIGENCE	AD HOC, AS NEEDED IMMEDIATELY BEFORE AND DURING BATTLE. LOW TRUST IN THE QUALITY, RELIABILITY AND RELEVANCE OF INTELLIGENCE. MOST DECISIONS ARE BASED ON PERSONAL AND INTUITIVE JUDGMENT.	INTELLIGENCE IS USED EXTENSIVELY BEFORE AND DURING BATTLE. TRUST IN ITS QUALITY, RELIABILITY AND USEFULNESS IS MEDIUM TO HIGH. DECISIONS ARE PARTLY PERSONAL BUT INCREASINGLY DEPEND ON INTELLIGENCE MUCH LESS ON INTUITION. INTELLIGENCE CAN BE IGNORED OR OVERRULED BY PERSONAL DECISIONS BUT INCREASED RELIANCE ON INTELLIGENCE INPUT.	MAJOR DECISIONS WITHOUT INTELLIGENCE ARE UNLIKELY. GREAT DEPENDENCE ON INTELLIGENCE AND MUCH LESS RELIANCE ON PERSONAL JUDGMENT. "ADDICTION" TO INTELLIGENCE MAKES INDEPENDENT-CREATIVE AD HOC DECISIONS LESS LIKELY. "COLLECTIVE DECISIONS" TEAM AND STAFF WORK. HEAVY RELIANCE ON INTELLIGENCE.
ORGANIZATION	NO PERMANENT ORGANIZATION. INITIATED BY THE LOCAL REQUIREMENTS OF THE COMMANDER.	LARGE PERMANENT INTELLIGENCE ORGANIZATIONS AND BUREAUCRACY, PRIMARILY MILITARY. RELATIVELY SMALL IN PEACE-TIME, GREATLY EXPANDED IN WAR.	VERY LARGE AND PERMANENT INTELLIGENCE ORGANIZATIONS AND BUREAUCRACIES, BOTH MILITARY AND CIVILIAN.
SCOPE & RANGE	MOSTLY SHORT-RANGE "LINE OF VISION" DIRECT OBSERVATION. PRIMARILY TACTICAL/OPERATIONAL.	SHORT & LONGER-RANGE. ENTIRE THEATER OF WAR-TACTICAL, OPERATIONAL AND STRATEGIC	GLOBAL AND THEATER ON ALL LEVELS.
SOURCES AND RELIABILITY	HUMINT - SPIES, DIRECT OBSERVATION. DIFFICULT TO TRANSMIT IN REAL TIME. LOW RELIABILITY, HENCE LITTLE TRUST IN ITS VALUE.	ALL SOURCES - HUMINT, PHOTINT, SIGINT CAN READILY BE TRANSMITTED IN REAL TIME. RELIABILITY & TRUST VARIES DRASTICALLY ONCE BATTLE BEGINS, DEPENDING ON CIRCUMSTANCES.	ALL SOURCES AND ALL LEVELS - HUMINT, PHOTINT, SIGINT CAN BE TRANSMITTED IN REAL TIME. AN INCREASED CAPACITY FOR THE COMMANDER TO COLLECT HIS OWN RELIABLE INTELLIGENCE IN BATTLE.

FIGURE 2 CONTINUED

	1800*	1914-1945	PRESENT AND FUTURE
PROBLEMS	LACK OF TRUST, LACK OF INTEREST. UNCERTAINTY CREATED BY DEARTH OF INTELLIGENCE (UNCERTAINTY OF TYPE A DOMINATES)	OFTEN POOR UNDERSTANDING OF WHAT INTELLIGENCE CAN AND CANNOT DO, TRUST IS BASED ON PAST PERSONAL EXPERIENCES. ATTITUDE STILL SHAPED BY DOCTRINES FORMULATED IN PREMODERN TIME (I.E. 1800 S)	GREATER DEPENDENCE ON INTELLIGENCE, THOUGH NOT ALWAYS BETTER UNDERSTANDING. OVERRELIANCE ON INTELLIGENCE WHILE ART OF COMMAND AND WAR ARE NEGLECTED. UNCERTAINTY CREATED BY TOO MUCH INTELLIGENCE (UNCERTAINTY TYPE B DOMINATES)
THE BALANCE OF INTELLIGENCE	ALL SIDES FACE EQUAL PROBLEMS.	SOME SIDES ARE BETTER OR MUCH BETTER IN INTELLIGENCE GATHERING. ADVANTAGE CAN SHIFT PERMANENTLY OR TEMPORARILY FROM ONE SIDE TO ANOTHER. INTELLIGENCE BECOMES A MAJOR WEAPON OR FORCE MULTIPLIER/DIVIDER.	AS IN 1914-1945, BUT THE GAP BETWEEN THE BEST AND WORST IS WIDENED. THE OPTIMAL USE OF MODERN PRECISION-GUIDED WEAPONS REQUIRES MORE AND BETTER REAL-TIME INTELLIGENCE.
SOLUTIONS TO PROBLEMS AND BETTER USE	GREATER RELIANCE ON THE EXPERIENCE AND INTUITION OF THE COMMANDER, THE ART OF WAR. TRANSFERENCE OF UNCERTAINTY BY INITIATIVE, MOVEMENT & OFFENSIVE TO OTHER SIDE. MAXIMUM CONCENTRATION, KEEPING RESERVES AT HAND. "STAND LIKE A ROCK". INTUITIVE RISK-TAKING.	BETTER UNDERSTANDING OF THE VALUE AND LIMITS OF INTELLIGENCE; POSITIVE EXPERIENCE. ART OF WAR AS IN 1800 THOUGH INCREASED RELIANCE ON INTELLIGENCE. CALCULATED RISK-TAKING.	NEED FOR BETTER EDUCATION IN INTELLIGENCE MATTERS. RELEARNING & FORMULATING THE ART OF WAR IN RELATION TO INTELLIGENCE. NEED TO KNOW WHEN TO TAKE ACTION AND NOT WAIT FOR THE LAST BIT OF INTELLIGENCE. MORE DEDICATED REAL TIME INTELLIGENCE. BETTER ORGANIZATION PREPARATION DISTILLATION AND DISTRIBUTION OF INTELLIGENCE.

* OCCASIONALLY, COMMANDERS LIKE WELLINGTON AND SHERMAN PAY GREAT ATTENTION TO INTELLIGENCE. SUCH MARGINAL IMPROVEMENT CAN UPSET THE BALANCE OF INTELLIGENCE IN ONE SIDE'S FAVO R AND HAVE MAJOR OPERATIONAL CONSEQUENCES.

1945 as it did between 1815 and 1914. Since 1945, armies have become increasingly bureaucratized. In business, bureaucratization kills the entrepreneur. In war it kills the general. Clausewitz wrote about the characteristics and role of just one type of general, the field commander in battle. Today the perfect general also requires other traits; he needs to have a genius for logistics, intelligence, military diplomacy, the management of a large planning staff and machine.[63] No single commander or military system may be able to develop all of these traits fully and simultaneously – indeed, strength in one may necessarily cause weakness in another. Thus the German Army may have been so bad at strategy precisely because it was so good at operations, just as the British Army lost every battle but the last. Whether those rules which govern field command hold true for battle management is unclear, but one thing is clear: this issue will be fundamentally affected by the nature of intelligence in the age of the bureaucratization of war. Since 1945, armies have massively increased their ability to acquire information, but this has been mostly opaque in nature, requiring complex analysis to be of use. The ability of armies to analyse such material with accuracy and speed has not kept pace with the increase in its volume; meanwhile, staffs have expanded in size and the quantity of their paperwork has been multiplied by their ability to communicate in real-time.

In particular, modern command, control, communications and intelligence systems have revolutionized the art of operations and the nature of generalship. One can acquire reliable information on issues which, for Clausewitz, were highly problematical, such as the location and strength of one's own forces. Even mediocre intelligence services can reduce the depth of the fog of war regarding the enemy's location and strength. Good intelligence services can almost entirely eliminate the fog on such issues, and even fathom the enemy's intentions before they are executed. In Clausewitz's theory of operations, intelligence and prolonged calculation handicapped command. In this century they have become a prerequisite for commanders.

Meanwhile, the relation between time and space in operations has changed. In Clausewitz's theory, and in his day, communications were slow, the speed of movement was low and operational theaters were relatively small. A battle could be affected by forces within just a day's march of one's forces, or roughly 20 miles. Of course, on the intersection between the operational and strategic levels, when planning major operations or the beginning of a campaign, one had to account for the forces in an entire country. In this century, real-time communication exists between forces scattered over millions of square miles. The increased duration of operations in time and their expansion in space, combined with the greater speed of movement of military forces, dramatically increased the size of the area from which enemy forces may affect a battle. For Clausewitz, the area of battle and the theater of operations overlapped in a relatively small space. The tactical and the operational spheres were almost identical. In this century, increasingly the tactical, operational and strategic spheres have come to overlap. MacArthur's battles could be immediately affected by forces hundreds of miles away, his operations by enemies based thousands of miles away. For commanders in the 1990s, this situation has become even more notable.

Hence, contemporary commanders may face a situation unprecedented in history. Intelligence and communications have improved, but so have the speed of battle and the need for quick decisions. More information is available more rapidly on more subjects; the one thing which has not changed is the speed required to make human judgements and decisions. Commanders need far more information on a far greater range of matters than in the past. They can also acquire it. Once, most pieces of intelligence were false; now, they are true, but trivial in quality and overwhelming in quantity. More can be worse. Certainly, a massive volume of accurate but opaque intelligence will produce information overload and two opposite problems. The first is the Clausewitzian dilemma of ignorant uncertainty, bewilderment and paralysis of the will. The second is self-confident error, the tendency to interpret this otherwise unmanageable mass of material solely through preconception. The second problem renders an increased quantity of intelligence counter-productive. Virtually any pre-conception will plausibly explain some intelligence. Hence, an increased quantity of opaque intelligence will support any and every mistake in assessment – volume reinforces error. The greater the quantity of such material, the greater the amount of intelligence any preconception will seem to explain and therefore, intuitively, the more accurate it will seem to be.

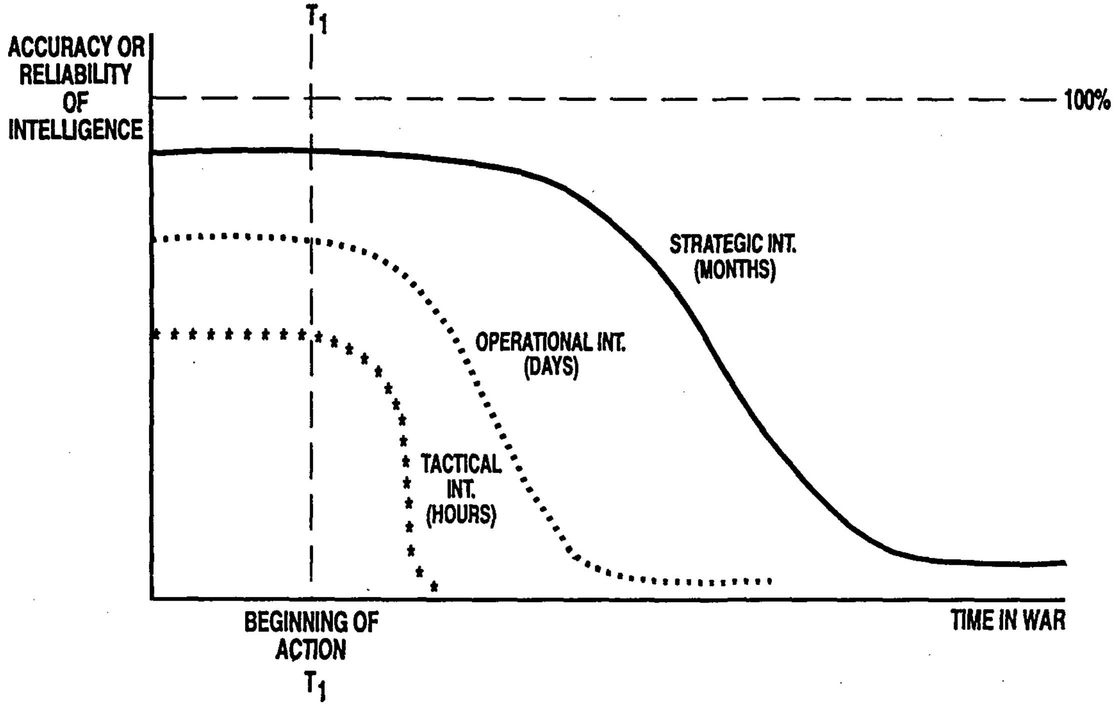
Simultaneously, the very nature of 'uncertainty' has changed. Once it stemmed simply from the lack of accurate knowledge; now both from the lack of accurate knowledge and of usable knowledge. In fact, contemporary commanders confront two distinct varieties of uncertainty, which have different causes and different solutions. One of them is as old as war, the other is the direct product of an attempt to solve the first. *'Type A' uncertainty* is that described by Clausewitz, characterized by ignorance and the inability to receive accurate, useful and timely intelligence in time to act on it. *'Type B' uncertainty* stems from the power of modern C^3I systems, which collect anything and everything that can be collected and communicate anything and everything that can be communicated simply because it can be done. In modern war, too much has to be known too quickly and too much can be known too often; and by the time what can be known has been understood, new factors become known which, again, have to be understood before one can act, ad infinitum. *Type A uncertainty* was literal and perceptual: (a) one could not know the facts and (b) one knew that fact. With *Type B*, the literal problem has declined while the perceptual one has changed: (a) sometimes one can know the facts and (b) one knows that one can do so but (c) one cannot predict how or when one will know the facts one wants to know or determine how to separate them in time from the mass of irrelevant facts that surround them – or distinguish 'signals' from 'noise', to use Roberta Wohlstetter's classic terminology.[64] *Type B uncertainty* produces the 'Schwarzkopf syndrome', the desire to wait just one more moment in order to read just one more report, the reluctance to act on imperfect knowledge because it is known to be imperfect and that at any point another report might well produce perfection. Clausewitz offered an elegant and practical solution to *Type A uncertainty*: to ignore intelligence and to act exclusively on informed intuition, because only through this means could generals hope to deal with the world. Neither this solution nor a simplistic application of Clausewitz's '*imperative principle*' can appeal to commanders confronting the second sort of uncertainty.

All this is affected by yet another problem. Pieces of intelligence are a perishable commodity since their value declines over time. (See Figure 3.) Two issues define the 'half-life' of military intelligence, or the speed with which it must be used in order to be of value: the level of war (tactical, operational or strategic) and the fluidity of the engagement. Generally speaking, the higher the level, the longer the half-life. There is, however, one fundamental limit to this generalization. The central issues at hand are the degree to which a military situation is constant and a piece of intelligence is unique. The more continual the situation and the more continuous the information, the longer the half-life; the less the continuity, the shorter the period of value. The half-life of the intelligence available to a battalion before the start of a prepared battle may match that of a corps midway through a mobile engagement. The tactical intelligence available to Canadian companies before the assault on Vimy Ridge in 1917 remained useful for weeks. The operational intelligence available to Rommel's headquarters during the 'Crusader' operation of 1941 often had a half-life of half a day.

The rate of decline in this 'half-life' from higher to lower levels varies directly with the fluidity of the engagement. This rate is most gradual in

FIGURE 3
INTELLIGENCE AS A PERISHABLE COMMODITY IN WAR

INTELLIGENCE AS A PERISHABLE COMMODITY IN WAR



set-piece battles, fought from fixed positions after days to months of preparation. In such circumstances, much of the deployment of enemy forces will remain stable for long periods and can easily be determined, while one's own chain of command and flow of intelligence will remain strong. Intelligence services will tend to function at their highest level of efficiency, producing and cross-checking useful, reliable and accurate intelligence. Staffs will have ample time to assess this intelligence, which will let the highest and best organized of them guide their subordinates. Often, neither the purport of intelligence nor the military situation will change notably from day to day. A broad framework of things constant and known will surround even those things which change and become unknown. In prepared operations, finally, whether one acts on a piece of intelligence immediately or two weeks later will as often as not be irrelevant. In many cases, actions taken immediately will not be completed for two weeks. In such circumstances, the half-life of tactical intelligence will be measured in days or weeks, operational intelligence in weeks or months, and strategic intelligence in months.

In mobile war, none of these rules holds true. The deployment of one's forces and those of the enemy will frequently change. So will intelligence about them. The chain of command and the flow of information will often break; intelligence and operational authorities at higher headquarters will not be able to guide their subordinates, which will often have to act with little time for assessment or preparation. Hence, in mobile war, the lower the level of the engagement, the less the accuracy of the intelligence and the power of assessment; and yet the greater the speed with which intelligence must be used to be of value. Therefore, *in mobile war the half-life of intelligence declines sharply*; from months at the strategic level to days at the operational and hours at the tactical level. Intelligence has the shortest half-life of all in the middle of a fluid engagement at the tactical sphere. Here, military situations and military intelligence are most fleeting and unique. In such circumstances, *intelligence literally must be used within a few hours or never at all*; often one must act immediately in ignorance rather than wait an hour for accurate information to arrive. At the operational level, the half-life of intelligence is greater, but nowhere else does assessment confront such technical complexity. Assessing staffs are small while irrevocable decisions often must be made within the space of several hours.

Each of these combinations of level and fluidity of war produces distinct problems regarding the collection, assessment and use of intelligence. Each of these problems has a different ideal solution. The lessons learned from any one need not apply to any other – in fact, they may be precisely the solution not to apply. Thus, elaborate intelligence organizations and careful calculation are essential for prepared operations, but counter-productive for fluid ones; similarly, experience of tactical command in a mobile war cannot prepare one for the accuracy and half-life of intelligence available at the strategic level.

These issues are not merely of academic significance. They are fundamental to command decisions and military education. Mere intellectual knowledge of them will not make good generals, but ignorance will make bad ones. Such knowledge cannot tell generals how they should act. It will show them the environment in which their actions take place: it will illuminate the problem and indicate the solutions. Perhaps commanders can be taught to recognize that the relationship between intelligence, time, uncertainty and decision takes different forms, each with distinct characteristics and problems. Perhaps new rule-of-thumb solutions can be found to these problems, through focusing on the collection of transparent intelligence, or on kinds of opaque intelligence which cannot be explained by every kind of preconception – by developing more effective means of formal assessment or schooled intuition and by formulating a version of the art of war resting on a higher base of knowledge. Perhaps one can define indicators which are most likely to reveal what one needs to know at a time that one can act on it. Perhaps a new version of the '*imperative principle*' can be formulated to help commanders know when they know enough to act, to know when they should cease to wait for new intelligence reports and disregard the constant flood of incoming trivia. Perhaps the main requirement of determination will be the ability to know that the time has come to 'stand like a rock' and move – despite the sure knowledge that new and relevant information will arrive while one is doing so. When one can know everything, the problem is to know when one knows enough. The dilemma is clear. Once commanders had to be taught to consider intelligence. Now that they have begun to do so, they must be taught how to overrule it. Perhaps only a 'military genius' will be able to deal with these new circumstances – but, if so, this genius will have different characteristics from those defined by Clausewitz.

Whatever proves to be the cause in this realm, more definite statements may be made about the effect of all these issues on prevailing theories of strategy and generalship. In wars of the future, intelligence just as much as uncertainty will shape what commanders will know and how they should act. War will remain an act of violent, frightening and dangerous competition, characterized by uncertainty of old kinds and new; only remarkable traits of character and mind can come to terms with it. Yet the nature of that uncertainty will not necessarily be the same as in Clausewitz's time. In a world of real-time communications, an abundance of intelligence and, often enough, a reduced number of known unknowns, the nature of the military genius will change. In the age of military machines, the personal traits of the commander will be less fundamental than in Clausewitz's day, although still important. He will have to know how to deal with both types of uncertainty. The nature of his intuition, his ability to understand intelligence and to act will be different.

In particular, the nature of military genius varies with circumstances. No single and universal mixture of characteristics produces this state at all times through the ages; in some circumstances military genius is less necessary for success in the field of battle than others; in others it may in fact be unnecessary or counter-productive. When intelligence reduces the reign of uncertainty, such personality characteristics as caution, daring and determination are transformed: they enter an alloy with knowlege, which alters many of the characteristics of the original element. Hence, one must take care in comparing the 'Great Captains' of this age with those of earlier epochs, because the greater degree of knowledge available to captains has fundamentally changed the characteristics which make them great. Bonaparte's determination literally is Haig's obstinacy. Intelligence allows far more commanders to achieve the status favored by Clausewitz, where boldness is 'governed by superior intellect'. It also allows caution to become deliberate – to become calculation.

All this, in turn, affects fundamental aspects of Clausewitz's theory of war. Whereas Jomini compared war to a game of chess, Clausewitz argued that 'in the whole range of human activities, war most closely resembles a game of cards'.[65] But which game? With what range of chance and what room for skill and what degree of uncertainty? Vingt-et-un? Poker? Whist? Clausewitz also saw war as an act not of reason but of will, where intellect matters less than character. Daring and determination were more valuable to generals than knowledge; greater military consequences were produced by marginal changes in personality than in brain cells.[66] The characteristics required of a commander were those of an expert at poker rather than a master of chess. Yet in the modern era war can also be an act of intelligence. When this is so, character is no longer by necessity the only central issue, and the nature of the metaphorical game itself has changed. The realm of chance is changed and the room for skill increased.

Commanders are no longer playing poker, but games where the realm of uncertainty is reduced, such as happens to different degrees in chess or whist. Nothing can be more fatal than to play chess or whist as though they were poker against an opponent who plays by the rules of the game. In particular, when intelligence lets one side play chess against the enemy's poker, caution ceases necessarily to be the mark of the undistinguished mind or the mere product of psychological weakness, whereas aggressive risk-taking may well be so. Caution can cease to be just a personality reflex. It can become a rational response to knowledge. A calculated chess player will have an advantage over a aggressive one who plays the game by the rules of poker. One cannot win at poker without bluffing; one cannot bluff at chess.

All this modifies Clausewitz's insights into the art of command. In principle, boldness – both in his sense of the word or in the common parlance of aggressive risk-taking – can become more common and more successful than he predicted. Yet deliberate caution may also become more common and even more effective than Clausewitz predicted. To err on the side of caution is no longer necessarily to err, for caution as well as boldness can be linked with effect to superior intellect or to cunning. The rise of intelligence and the decline of uncertainty in modern warfare have affected the actions of generals. They must also affect our judgement of generalship.

NOTES

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1. Carl von Clausewitz, *On War* (eds Michael Howard and Peter Paret, Princeton, 1976), p.233.
2. Ibid. p.117. For a discussion of many issues related to this topic, see Michael Handel, 'Intelligence and Military Operations' in Michael I. Handel (ed.), *Intelligence and Military Operations* (London, 1990) and *Masters of War: Sun Tzu, Clausewitz and Jomini* (London, 1992).
3. W. Kendrick Pritchett, *The Greek State at War, Part III: Religion* (Berkeley, CA, 1982), p.149.
4. Clausewitz, *On War*, pp.198–201, 210, and Handel, *Military Operations*, pp.8–11, 13–21 and *Masters of War*, pp.107–32.
5. For Jomini, see Handel, *Masters of War*, pp.107–32. For recent accounts of Wellington and operational intelligence, cf. Jac Weller, 'Wellington's Asset: A Remarkable Successful System of Intelligence', *Military Review*, 42 (June 1962); Julia V. Page, *Intelligence Officer in the Peninsula: Letters and Diaries of Major the Hon. Edward Charles Cocks, 1786–1812* (Tunbridge Wells, 1986: introduction by David Chandler). All modern accounts of Wellington's military organization also refer to his intelligence system: cf. Michael Glover, *Wellington's Army in the Peninsula, 1808–1814* (Vancouver: 1977); H.C.B Rogers, *Wellington's Army* (London, 1979) and Lawrence James, *The Iron Duke, A Military Biography of Wellington* (London, 1992). Stephen Mark Harris's MA dissertation at the Department of History,

University of Calgary, 1993, 'British Military Intelligence in the Crimean War, 1854–1856', demonstrates that at their best, operational intelligence systems in the middle nineteenth century could provide material which rivals the quality and accuracy of their successors of the twentieth century.

6. Handel, *Masters*, pp.102–7.
7. Clausewitz, *On War*, p.100.
8. Ibid., p.110.
9. Ibid., p.103.
10. Ibid., p.101.
11. Ibid., p.108.
12. Ibid., pp.105–6.
13. Ibid., p.117.
14. Ibid., pp.102–3.
15. Ibid.
16. Ibid., p.102.
17. Ibid., pp.146–7.
18. Ibid., p.71.
19. Ibid., p.112.
20. Ibid., p.108. (Our italics.)
21. Ibid., pp.190, 192.
22. Ibid., p.200.
23. For arguments on these lines, cf. 'Intelligence and the Problem of Strategic Surprise', in Michael Handel, *War, Strategy and Intelligence* (London, 1989), pp.229–81 and the same author's *Perception, Deception and Surprise: The Case of the Yom Kippur War* (Jerusalem, 1976); Richard K. Betts, 'Analysis, War and Decision: Why Intelligence Failures are Inevitable', *World Politics*, 31 (Oct., 1978), pp.61–89, and the same author's *Surprise Attack: Lessons for Defense Planning* (Washington, 1982).
24. Clausewitz, *On War*, pp.110, 190–1; Azar Gat, *The Origins of Military Thought, From the Enlightenment to Von Clausewitz* (Oxford, 1989), pp.202, 205.
25. Clausewitz, *On War*, p.192.
26. Ibid., pp.202–3.
27. Ibid., p.85.
28. Ibid., p.180. See also Handel, *Masters of War*, pp.151–4.
29. Ibid., pp.103, 190.
30. Ibid., p.86.
31. Michael Howard, 'The Influence of Clausewitz', pp.29–39 in Howard and Paret, op. cit. For contemporary views about determination, character and daring similar to those of Clausewitz cf. Marshall Marmont, *The Spirit of Military Institutions* (trans. Frank Schaller), (1974 (1864)), pp.222–37. Clausewitz's great nineteenth-century competitor as a theorist of strategy, the Baron de Jomini, placed far less emphasis on the topic but his few references to the issue coincide with the views of Clausewitz. See Baron de Jomini, *The Art of War*, translated by G.H. Mendell and W.P. Craighill, (Philadelphia, 1962), p.50. The Clausewitzian and Napoleonic tradition about the psychology of command also pervades the views of one of the most representative commentators on strategy before the First World War, Commandant J. Colin, *The Transformations of War*, translated by L.H.R. Pope-Hennessey (London, 1912), pp.183, 351–3.
32. John Keegan, 'Peace By Other Means: War, Popular Opinion and Politically Incorrect Clausewitz', *The Times Literary Supplement*, 11 Dec. 1992, pp.3–4.
33. Martin Blumenson, *The Patton Papers, 1885–1940* (Boston, 1972), pp.753, 794–7 and *The Patton Papers, 1940–1945* (Boston, 1974), p.263; Alfred D. Chandler, Stephen E. Ambrose, Joseph P. Hobbs, Edwin Alan Thompson and Elizabeth F. Smith (eds), *The Papers of Dwight David Eisenhower III: The War Years* (Baltimore, 1970), pp.1439, 1754. See also Christopher Brassford, *Clausewitz in English: The Reception of Clausewitz in Britain and America, 1815–1945* (New York, 1994), especially pp.152–87.

34. John Keegan, *A History of Warfare* (New York, 1993), simultaneously views Clausewitz as having had no influence while somehow still producing every evil in modern war. Unfortunately, he fails to understand the text he attacks, describing a Clausewitz who never lived while ignoring the one who did. Keegan's appreciation of Clausewitz is reminiscent of the Ayatollah Khomeini's views on the United States.
35. Brian Bond, *Liddell Hart: A Study of His Military Thought* (London, 1977), pp.20, 45; John Mearsheimer, *Liddell Hart and the Weight of History* (Ithaca, 1988), pp.48–50; Blumenson, *The Patton Papers, 1885–1940*, pp.794–7.
36. For a discussion of this issue, cf. John Ferris, 'The Intelligence-Deception Complex: An Anatomy', *Intelligence and National Security*, Vol.4, No.4 (October, 1989).
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